

UNIVERSIDAD POLITÉCNICA DE MADRID

ESCUELA TÉCNICA SUPERIOR DE INGENIEROS DE TELECOMUNICACIÓN



MASTER'S DEGREE IN SIGNAL THEORY AND
COMMUNICATIONS (MSTC)

SPECIALISATION: SIGNAL PROCESSING AND MACHINE LEARNING FOR BIG DATA

MASTER'S THESIS

SUBJECT-INDEPENDENT CLASSIFICATION
OF PARKINSON'S DISEASE FROM
FREQUENCY-MODULATED CONTINUOUS
WAVE RADAR MICRO-DOPPLER GAIT
SIGNATURES

ZIAD BOUSSEDRA

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Title: Subject-Independent Classification of Parkinson's Disease from Frequency-Modulated Continuous Wave Radar Micro-Doppler Gait Signatures

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RESUMEN

La enfermedad de Parkinson es el segundo trastorno neurodegenerativo más frecuente, y la dificultad para caminar figura entre sus signos más tempranos e incapacitantes. El radar de onda continua modulada en frecuencia (FMCW) permite observar la marcha sin contacto, sin cámaras y sin sensores corporales: la grabación es una **firma micro-Doppler**, una imagen tiempo-frecuencia que integra el movimiento de todas las partes del cuerpo y de la que pueden extraerse las velocidades del torso y de las extremidades. Por ello el radar resulta atractivo para evaluar el movimiento en la clínica y en el hogar; sin embargo, hasta donde sabemos, ningún trabajo publicado había evaluado un clasificador de Parkinson frente a controles sobre estas firmas con **validación independiente del sujeto**, aquella en la que cada persona puntuada es alguien que el modelo nunca vio durante el entrenamiento, que es la condición real de una herramienta de cribado.

Este trabajo construye esa evaluación sobre una red de radar clínicamente validada y una cohorte de pacientes de Parkinson y controles sanos. Se desarrolla un proceso completo de análisis y se evalúan desde modelos clásicos hasta varias redes neuronales profundas, incluida la arquitectura de referencia de la literatura, con validación cruzada dejando un sujeto fuera; cada modelo se compara con referencias sencillas, como un predictor que solo conoce la duración de cada grabación, y cada resultado se acompaña de su intervalo de confianza.

Bajo esta evaluación, ningún clasificador demuestra conocer la enfermedad más allá de la duración de la grabación: los pacientes sencillamente tardan más. El análisis de atención atribuye la puntuación restante a propiedades de cada grabación y no a la marcha; y reproducir sobre los mismos datos el reparto aleatorio habitual en la literatura muestra cuánto infla ese diseño el rendimiento: modelos aparentemente listos para la clínica se desploman ante desconocidos. Lo que sobrevive es clínicamente coherente: el radar mide con fiabilidad los tiempos de la marcha, una prueba *Timed-Up-and-Go* automatizada.

La contribución es doble: la primera evaluación independiente del sujeto de este problema, y una demostración de cómo los clasificadores de marcha por radar llegan a parecer mejores de lo que son, junto con el protocolo de evaluación que lo impide.

SUMMARY

Parkinson's disease is the second most common neurodegenerative disorder, and difficulty walking is among its earliest and most disabling signs. Frequency-Modulated Continuous Wave (FMCW) radar can observe a walk without contact, cameras or wearable sensors: the recording is a **micro-Doppler signature**, a time-frequency image that integrates the motion of every part of the body, from which the velocities of the torso and the limbs can be extracted. Radar is therefore attractive for assessing movement in clinics and homes, yet, to the best of our knowledge, no published work had evaluated a Parkinson's-versus-control classifier on these signatures under **subject-independent**

validation, in which every scored person is one the model never saw during training, which is the condition a screening tool faces in practice.

This thesis builds that evaluation on a clinically validated radar network and a cohort of Parkinson's patients and healthy controls. A full pipeline is developed, and models from classical baselines to several deep neural networks, including the field's reference architecture, are evaluated with leave-one-subject-out cross-validation; every model is compared against simple baselines, such as a predictor that knows only how long each recording lasts, and every result carries a confidence interval.

Under this evaluation, no classifier demonstrates more knowledge of the disease than recording duration alone: patients simply take longer. Attention analysis traces the remaining score to properties of each recording rather than of the walk, and reproducing the literature's usual random hold-out on the same data shows how strongly that design inflates performance: models that look clinic-ready on familiar subjects collapse on strangers. What survives is clinically coherent: radar reliably measures the timing of the walk, an automated *Timed-Up-and-Go* test.

The contribution is twofold: the first subject-independent evaluation of this problem, and a demonstration of how radar gait classifiers come to look better than they are, together with the evaluation protocol that prevents it.

PALABRAS CLAVE

Enfermedad de Parkinson; radar FMCW; micro-Doppler; análisis de la marcha; aprendizaje profundo; validación independiente del sujeto; control de factores de confusión.

KEYWORDS

Parkinson's disease; FMCW radar; micro-Doppler; gait analysis; deep learning; subject-independent validation; confound control.

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Contents

Resumen y Palabras Clave	iv
Acknowledgements	vi
List of Acronyms	xi
1 Introduction and Objectives	1
1.1 Motivation	1
1.2 Problem Statement	1
1.3 Contributions	2
1.4 Structure of this Document	2
1.5 Objectives	3
2 State of the Art	5
2.1 Contactless gait sensing	5
2.2 Micro-Doppler foundations and gait validation	5
2.2.1 The source of the dataset used in this thesis	6
2.3 Deep learning for micro-Doppler gait classification	6
2.4 Learning from small radar datasets	7
2.5 Radar for neurodegenerative disease and Parkinson's	7
2.6 Limitations of the state of the art in the field	8
2.7 Summary of the most influential works	8
2.8 Research gap and positioning of this thesis	10
3 Materials and Methods	11
3.1 Dataset	11
3.1.1 Provenance	11
3.1.2 Acquisition protocol	12
3.1.3 Cohort	13
3.1.4 Contents of one recording	13
3.1.5 Quality control	14
3.2 Radar System and Signal Model	14
3.2.1 From a transmitted chirp to a spectrogram	14
3.2.2 Doppler shift and velocity	16
3.2.3 What a single spectrogram column represents	17
3.2.4 The two analysis bands	17
3.2.5 Structure visible in a single recording	17
3.2.6 Coupling between the two channels	18
3.3 Preprocessing	19
3.3.1 Windowing	21
3.3.2 Fold-local standardisation	22
3.3.3 Resampling and channel stacking	23

3.3.4	Window weighting	23
3.3.5	Direction-of-travel labelling	24
3.4	Handcrafted Features	25
3.4.1	Behaviour with respect to recording length	25
3.4.2	Features chosen for clinical reasons	26
3.5	Statistical Analysis	27
3.5.1	Rank-based comparison	27
3.5.2	Effect size	28
3.5.3	Correction for multiple comparisons	28
3.5.4	Subject-level similarity	28
3.6	Classical Baselines	29
3.7	Deep Models	31
3.7.1	SmallCNN, trained from scratch	32
3.7.2	ResNet-18 as a linear probe	33
3.7.3	Envelope-LSTM	34
3.7.4	Training configuration	34
3.8	Validation Protocol	35
3.9	Confound-Control Protocol	37
3.10	Evaluation Metrics	37
4	Results	39
4.1	Dataset Characterisation and Exploratory Analysis	39
4.1.1	Cohort composition	39
4.1.2	Recording duration differs between groups, and the chair is why . .	39
4.1.3	Nearly every feature differs, which is a warning rather than a result	41
4.1.4	A walk identifies the walker	42
4.1.5	How the exploratory analysis constrained the design	44
4.2	The Null Floors	44
4.3	Classical Baselines	45
4.4	Deep Models	46
4.5	Ablation Studies	47
4.6	Decision Performance at a Fixed Threshold	49
5	Discussion, Conclusions and Future Work	51
5.1	Discussion	51
5.1.1	The random-split demonstration: the gap is the protocol	51
5.1.2	The same demonstration with the field's own network	53
5.1.3	What the models actually read	55
5.1.4	The intervention test: removing what they read	57
5.1.5	What was found, stated positively	58
5.2	Limitations	59
5.3	Conclusions	59
5.4	Future Work	60
5.5	Closing Remarks	62
	Bibliography	63
	Annex A: Ethical, Economic, Social and Environmental Aspects	67
	Annex B: Economic Budget	69

List of Figures

3.1	Published acquisition environment	12
3.2	Acquisition geometry and signal chain	15
3.3	Control and PD recordings in both channels	18
3.4	Channel coupling at the heel strike	19
3.5	The deep-learning pipeline, end to end	20
3.6	Windowing of one recording	22
3.7	One cached window	23
3.8	Training influence per subject	24
3.9	Direction-of-travel labelling	25
3.10	One fold of the classical baseline	29
3.11	Model capacity on a logarithmic scale	32
3.12	SmallCNN, layer by layer	32
3.13	ResNet-18 as used here	33
3.14	One fold of the nested protocol	36
4.1	Cohort composition	39
4.2	Recording duration by group and variant	40
4.3	Effect size per feature	41
4.4	Group distributions of the strongest length-independent features	42
4.5	Subject fingerprint strength	43
4.6	The baseline grid as a heat map	46
4.7	The deep campaign against the floors	47
4.8	Confusion matrices at the threshold	49
4.9	Per-subject probabilities, sorted	50
4.10	One case per confusion cell	50
5.1	The collapse test	53
5.2	AlexNet: one architecture, two protocols	54
5.3	AlexNet's decisions, seen versus never-seen	55
5.4	Grad-CAM gallery, SmallCNN	56
5.5	Grad-CAM gallery, ResNet-18 probe	56

List of Tables

2.1	Key related works and their influence on this thesis	9
3.1	Composition of the dataset	13
3.2	Acquisition parameters and what each one controls	16
3.3	The stages of the deep-learning pipeline	21
3.4	Handcrafted features and their length behaviour	26

3.5	Clinically motivated features and their predicted direction	27
3.6	The three classical classifiers and their roles	30
3.7	The five feature sets of the classical baseline	31
3.8	Retraining depth for the pretrained network	33
3.9	What the source publication contributes to the design	34
3.10	Shared training configuration of all deep models	34
3.11	The three groups of one fold and their roles	36
4.1	Recording duration by group and variant	40
4.2	Design decisions revised by the exploratory analysis	44
4.3	The null floors	45
4.4	Classical baseline AUCs	45
4.5	Deep models under leave-one-subject-out validation	46
4.6	Exploratory input variants	48
4.7	Decision metrics at the prevalence threshold	49
5.1	The validation ladder	52
5.2	AlexNet under the two protocols	54
5.3	Background-suppressed results	57
5.4	The positive findings	58
5.5	Limitations, and what would resolve each	59
5.6	Future work: the path to stronger results	61
5.7	Future work: methodological improvements	62
B.1	Presupuesto económico.	69

List of Acronyms

ACGAN *Auxiliary-Classifier Generative Adversarial Network*

AUC *Area Under the (ROC) Curve*

CI *Confidence Interval*

CNN *Convolutional Neural Network*

FDR *False Discovery Rate*

FMCW *Frequency-Modulated Continuous Wave*

GAN *Generative Adversarial Network*

GDPR *General Data Protection Regulation*

Grad-CAM *Gradient-weighted Class Activation Mapping*

LOSO *Leave-One-Subject-Out (cross-validation)*

LSTM *Long Short-Term Memory*

PD *Parkinson's Disease*

ROC *Receiver Operating Characteristic*

STFT *Short-Time Fourier Transform*

SVM *Support Vector Machine*

TFM *Trabajo Fin de Máster (Master's Thesis)*

TUG *Timed-Up-and-Go (test)*

1. Introduction and Objectives

1.1 Motivation

Parkinson's disease is the second most common neurodegenerative disorder, and its diagnosis remains clinical: a neurologist examines the patient's movement against consensus criteria [1]. Among the motor signs, **gait** holds a special place. It deteriorates early, it disables profoundly, and it is measurable: patients walk more slowly, with shorter steps, reduced arm swing and, characteristically, greater stride-to-stride variability and marked difficulty with turns and with rising from a chair [2, 3]. A technology that measured gait objectively, repeatedly and outside the clinic could support earlier referral, longitudinal tracking and therapy adjustment.

The instruments available today each concede something. Optical motion capture is the laboratory gold standard but is expensive, confined to instrumented rooms and requires markers. Wearable sensors travel with the patient but must be worn, charged and accepted, a real burden for elderly users. Cameras are cheap but raise immediate privacy objections in the home. **Radar** concedes none of these: it is contactless, works through clothing and in darkness, records no camera image, and its micro-Doppler return integrates the motion of every moving part of the body into one time-frequency image, the **micro-Doppler signature**, from which the velocities of the torso and the limbs can be extracted [4, 5]. In that image the torso traces a slow band and each foot a fast periodic arch.

This thesis works on data from a clinically validated multi-node FMCW radar network built precisely for gait monitoring, whose parameter extraction has been validated in both healthy and Parkinson's populations [6]. The cohort contains 58 subjects, 33 healthy controls and 25 Parkinson's patients, each walking a short corridor in a protocol modelled on the clinical Timed-Up-and-Go test. The question this thesis asks of that data is the natural next one: can the disease itself be read from these signatures?

1.2 Problem Statement

Radar has already proven itself around gait, but on different questions. One line of work **measures** gait: it validates which biomechanical parameters, such as step time, cadence and stride length, can be extracted reliably from the radar return [4, 6]. Another line **classifies** walkers into coarse groups, almost always young versus elderly adults, and reports accuracies above 90 % [7, 8]. Between the two sits the clinical question this thesis takes up: whether Parkinson's disease itself can be recognised in the radar signature of a walk, tested the only way a screening tool is used in practice, on people the classifier has never seen. To the best of our knowledge, no published work answers it.

How a classifier is tested decides what its score means, and this is where the field is weakest. A micro-Doppler recording carries a strong personal signature: published systems recognise individual *people* from their walk [9, 10]. If recordings of the same person appear in both the training and the test data, a classifier can therefore score highly by recognising the person and recalling their label, having learned nothing about the disease. The field’s own record shows the danger in another form: a network trained on micro-Doppler images and reaching 97.8% accuracy was shown by its own authors to have keyed on background differences between recording sites rather than on gait [7]. Other regularities offer the same shortcut, because Parkinson’s cohorts are typically older than their controls, and patients take longer to complete a walking test.

The problem this thesis addresses is therefore twofold. First, to build and evaluate a Parkinson’s-versus-control classifier on radar micro-Doppler signatures with every scored person excluded from training. Second, to do so under an evaluation design in which every score is defensible: each confound is measured, given an explicit baseline score that any real model must beat, and controlled rather than ignored.

1.3 Contributions

The thesis makes five contributions:

- **The first subject-independent evaluation, to our knowledge, of Parkinson’s-versus-control classification from radar micro-Doppler gait signatures.** The evaluation covers classical models and four deep model families, including AlexNet, the architecture behind the field’s best-known result [7].
- **A confound-aware evaluation protocol.** Every model is trained and scored with leave-one-subject-out validation on fixed-length windows, is judged against deliberately simple baseline predictors, such as one that knows only how long each recording lasts, and is reported with a subject-level confidence interval.
- **An honest measurement of how much signal the data holds.** Under this subject-independent evaluation, no classifier can be distinguished from the duration-only baseline. The finding is presented as a property of the data, with its uncertainty, not as a modelling failure.
- **An explanation of where the scores come from.** Attention analysis shows what the image networks actually read; an intervention experiment removes what they attend to; and a protocol experiment reproduces the literature’s numbers on this very dataset and shows them collapsing on unseen subjects.
- **A clinically usable finding and a reusable protocol.** Radar reliably measures how long the walk and its phases take, in effect an automated Timed-Up-and-Go test, and the evaluation protocol can serve as the design document for the richer datasets the field needs next.

1.4 Structure of this Document

The rest of this document is organised as follows:

- **Chapter 2, State of the Art:**

- Radar micro-Doppler as a contactless gait sensing modality;
- The classification studies closest to this task;
- The validation pitfalls that recur throughout the field.

- **Chapter 3, Materials and Methods:**

- The dataset and the acquisition system;
- Preprocessing, windowing and the extracted features;
- The classical and deep model families;
- The leave-one-subject-out protocol, the confound controls and the evaluation metrics.

- **Chapter 4, Results:**

- The exploratory characterisation that constrained the design;
- The baseline scores every model must beat;
- The classical and deep model campaigns and the input variants;
- Decision-level performance at a fixed threshold.

- **Chapter 5, Discussion, Conclusions and Future Work:**

- The experiments that show where the gap to the literature comes from;
- What the models actually read, and the intervention that tests it;
- The positive findings, the limitations and the conclusions;
- The path to stronger results.

- **Annexes:**

- Ethical, social, economic and environmental aspects;
- The project budget.

Code and reproducibility. The complete pipeline, the frozen pre-registered configuration, the analysis scripts that produce every figure and table, and the L^AT_EX source of this document are publicly available at <https://github.com/boussedraziad-sudo/parkinson-radar-thesis>. The clinical recordings themselves are not redistributed: they remain with the originating team under the ethics approval described in Annex A, and the repository documents how to run the pipeline on them.

1.5 Objectives

The general objective of this thesis is to determine whether Parkinson's disease can be classified from FMCW radar micro-Doppler gait spectrograms under an evaluation a clinician could trust. It is pursued through five specific objectives:

- **O1.** Characterise the dataset and identify the confounds capable of producing spurious classification performance, before any model is trained.
- **O2.** Establish a confound-controlled classical baseline under subject-independent (leave-one-subject-out) validation, including deliberately simple reference predictors, such as one that knows only how long each recording lasts, that set the score any real model must clearly beat.
- **O3.** Develop and evaluate deep models on windowed micro-Doppler signatures: a compact convolutional network, a pretrained ResNet-18 with a retrained final layer, and a sequence model on velocity envelopes, under the identical protocol.
- **O4.** Verify with interpretability methods (Grad-CAM) what the trained networks actually attend to, and test the finding with an intervention experiment rather than leaving it as a picture.
- **O5.** Report every result against the explicit null floors with honest uncertainty, and reconcile the outcome with the published literature by reproducing its validation design on this same dataset.

The conclusions of Chapter 5 return to these objectives one by one.

2. State of the Art

This chapter reviews the literature at the intersection of contactless radar sensing, micro-Doppler signal processing and machine-learning gait analysis, and positions the present work within it. The survey is organised from the physical substrate upward:

- The case for radar as a gait sensor (Section 2.1);
- What a radar micro-Doppler signature physically encodes, and how reliably it can be measured (Section 2.2);
- How deep learning has been applied to micro-Doppler signatures (Section 2.3);
- Strategies for learning from the small datasets typical of this field (Section 2.4);
- The emerging use of radar for neurodegenerative disease (Section 2.5);
- The validation pitfalls that recur throughout (Section 2.6).

Section 2.7 then condenses the most influential works into Table 2.1, and Section 2.8 distils the resulting research gap that this thesis addresses.

2.1 Contactless gait sensing

Quantitative gait analysis is clinically valuable for detecting and monitoring movement disorders, but each of the dominant instrumented modalities carries a cost. **Wearable inertial sensors** are accurate and inexpensive, yet they must be worn, which constrains continuous use and biases natural gait. **Optical motion capture** and video are accurate but expensive, illumination-dependent and privacy-invasive.

Radar has emerged as a complementary alternative that is **contactless**, insensitive to lighting and clothing, and **privacy-preserving**, which makes it attractive for continuous and in-home monitoring [4, 5, 11]. These properties motivate the growing body of work that uses radar to characterise human gait, reviewed in the remainder of this chapter.

2.2 Micro-Doppler foundations and gait validation

When a radar illuminates a walking person, different body parts moving at different radial velocities induce distinct Doppler shifts. The resulting time–frequency representation, the **micro-Doppler signature**, encodes the slow, steady motion of the **torso** at low Doppler frequencies and the faster swings of the **limbs** at higher frequencies [4].

Establishing which parts of this signature are physically trustworthy has been the subject of careful validation studies. Seifert et al. [4] benchmarked a 24 GHz Doppler radar

against marker-based motion capture and instrumented-treadmill ground-reaction forces. They showed that radar recovers **rhythm and pace** parameters (stride and step time, cadence, stride length) reliably, but that joint-velocity estimates and gait-event timing depend strongly on **geometry, viewing angle and walking speed**.

Complementing this, the survey of Gurbuz et al. [5] catalogues radar-based gait-parameter estimation for fall-risk assessment and notes a crucial limitation: radar captures cadence and step time well, but measures **gait variability** only poorly to moderately. This matters because stride-to-stride fluctuation is a defining sign of parkinsonian gait.

2.2.1 The source of the dataset used in this thesis

The radar network that produced the dataset analysed in this work is that of **López-Delgado et al.** [6], developed at the Universidad Politécnica de Madrid in collaboration with the Hospital General Universitario Gregorio Marañón de Madrid. That study sits squarely in the validation tradition described above: it compares multi-node radar configurations against a **Vicon** optical motion-capture reference [12] and demonstrates accurate spatiotemporal parameter extraction in both healthy and Parkinson’s populations.

Two of its findings directly motivate design decisions taken later in this thesis:

- **Foot-oriented radar nodes best recover temporal gait parameters** in motorically impaired subjects. This justifies treating the foot and torso channels as distinct, complementary information sources rather than merging them into one.
- **The torso-velocity/heel-strike coupling breaks down in Parkinson’s disease**, although it is reliable in healthy gait. This is a *relational* property of the signature: a two-channel image model can represent it, whereas scalar summary features cannot.

Critically, however, that work stops at parameter *measurement*: it contains **no disease classifier**. This is the point of departure for the present thesis, which takes the same acquisition system and asks whether the diagnosis itself can be predicted directly from the micro-Doppler signatures.

2.3 Deep learning for micro-Doppler gait classification

A second line of work treats the micro-Doppler signature as an image or a sequence, and applies machine learning to classify gait. Two architectural families dominate:

- **Convolutional networks (CNNs)** applied to the full time–frequency image.
- **Recurrent networks**, such as the **LSTM**, applied to one-dimensional velocity time-series, for example the upper, lower and power-weighted Doppler envelopes.

Hayashi et al. [7] exemplify the second family, classifying young versus elderly gait from 24 GHz micro-Doppler with a **long short-term memory** network on extracted envelopes. They reach **94.9 %** accuracy and report that the stance-phase (lower-envelope) feature alone was most discriminative. A companion simulation study [8] corroborates that **leg-motion features** carry most of the age-discriminative signal.

CNN-based approaches on micro-Doppler images are equally common across human-activity and gait recognition, with reported accuracies frequently in the 90–98 % range: Nguyen et al. [13] separate five gait patterns at 96 to 99 % with a millimetre-wave sensor, Ha et al. [14] reach 97.9 % with a compact CNN on cepstral features from six subjects, and Fard et al. [15] report 89.7 % under a leave-one-person-out protocol on three subjects. Notably, Papanastasiou et al. [9] and Ni and Huang [10] show that micro-Doppler encodes **individual identity** strongly enough to recognise people (>93.5 % and 96.7 % respectively). As Section 2.6 argues, that result has direct and cautionary implications for how disease classifiers must be validated. Alternative front-ends have also been explored, including audio-processing pipelines applied to the Doppler signature [14] and low-cost millimetre-wave sensing for gait [11].

2.4 Learning from small radar datasets

Radar gait datasets are typically small, on the order of tens of subjects, because data collection requires supervised sessions with specialised hardware. Two strategies recur for coping with this scarcity:

- **Transfer learning.** Seyfioğlu et al. [16] show that deep networks pre-trained on diversified micro-Doppler transfer effectively to new motion-classification tasks (97 % on a seven-class task against 86 % for the same network trained from scratch), and Park et al. [17] demonstrated transfer learning on micro-Doppler as early as 2016, reaching 80.3 % on five subjects.
- **Synthetic data augmentation.** Erol et al. [18] use a kinematically sifted auxiliary-classifier **GAN** to synthesise realistic radar micro-Doppler signatures and expand small training sets, raising cross-environment accuracy from 84 to 93 %.

Both strategies are directly relevant to a **58-subject** study, and they inform the transfer-learning and augmentation choices adopted in this thesis.

2.5 Radar for neurodegenerative disease and Parkinson’s

Applying radar gait analysis specifically to neurodegenerative disease is recent and still limited. The **STRIDE** system of Cai et al. [19] uses radar intelligence for Alzheimer’s-disease-and-related-dementia risk evaluation, illustrating radar’s promise for passive cognitive-decline screening.

For Parkinson’s disease specifically, machine-learning gait classification has been pursued mainly through **non-radar** modalities such as force sensors and wearables: Meral and Ozbilgin [20] report 99.4 % accuracy for Parkinson’s versus control from force-plate recordings under k -fold validation with no subject-independent test. The radar literature on Parkinson’s remains largely confined to parameter validation [6] rather than automated diagnosis. To the best of our knowledge, **no published work performs subject-independent Parkinson’s-versus-control classification directly from radar micro-Doppler signatures**. That is the specific gap this thesis targets. The accuracies reported across these works, gathered in Table 2.1, cluster between 90 and 99 %; the next section explains why they must be read with caution.

2.6 Limitations of the state of the art in the field

The high accuracies reported across the micro-Doppler classification literature must be read against a set of recurring methodological hazards, because several of them are known to inflate results. Four are relevant here.

- **Subject leakage.** Because micro-Doppler encodes individual identity so strongly [9, 10], any evaluation that allows recordings from the same subject to appear in both training and test sets permits the model to recognise *people* rather than the target condition. **Subject-independent (leave-one-subject-out)** validation is therefore essential, yet much of the field relies on random hold-out splits [7, 8]. Of the sixteen works reviewed in this chapter, **only two** report a genuinely subject-independent protocol [15, 17], and neither addresses Parkinson’s disease. Put differently, a walk varies in two ways: within a person from one recording to the next (intra-patient variability) and between people (inter-patient variability). A model evaluated on people it has already seen only has to master the first; the systems in this literature are never shown to model the second, which is why they fail on new patients.
- **Shortcut learning on artifacts.** CNNs trained on micro-Doppler images often latch onto irrelevant signals rather than gait. Hayashi et al. [7] explicitly demonstrate that an AlexNet reaching **97.8 %** did so by exploiting **background-noise differences**, which arose because their two groups were recorded at different locations. This failure mode is called **shortcut learning**: the network found an easier giveaway in the images, the noise pattern of each recording site, and used it instead of the gait. The same risk exists for any network trained on these images, which is why this thesis inspects, with attention maps, what its own networks actually look at.
- **The age confound.** Group-level confounds run through this whole literature, and **age** is especially dangerous for a Parkinson’s study because it is so easy to miss. Because micro-Doppler alone separates young from elderly walkers at **94.9 %** [7], and Parkinson’s cohorts are typically older than controls, a classifier can appear to detect disease while in fact detecting age.
- **Measuring what matters least reliably.** Among the features most diagnostic of parkinsonian gait, increased **stride-to-stride variability** is exactly what radar measures least reliably [5], and joint-velocity estimates depend strongly on geometry [4]. Pipelines that aggressively normalise amplitude, or that use very short temporal windows, therefore risk discarding the very pathology they seek.

Together these observations argue that **careful evaluation design** is the central unsolved problem in this niche, rather than raw accuracy. Concretely, that means subject-independent validation, explicit confound control, and interpretability checks confirming the model attends to physiologically plausible Doppler bands.

2.7 Summary of the most influential works

Table 2.1 condenses the studies that most directly shaped the design of this thesis, pairing each key finding with the concrete methodological consequence drawn from it.

Table 2.1: Key related works, their principal findings, and the specific methodological decision each one motivates in this thesis.

Work	Key finding	Influence on this thesis
López-Delgado et al. [6] (UPM, source of our dataset)	• Multi-node FMCW radar validated against Vicon. • Foot nodes best recover temporal parameters. • Contains no classifier.	• Provides the dataset and the geometry. • Motivates the two-channel input. • Defines the gap this work fills.
Hayashi et al. [7]	• LSTM on Doppler envelopes: 94.9 % young vs elderly. • Their AlexNet hit 97.8 % via a site artifact, not gait.	• Age treated as a dominant confound. • Attention checks made mandatory. • The strongest cautionary result.
Gurbuz et al. [5]	• Radar measures cadence and step time well. • Gait variability only poorly to moderately.	• Tempers expectations for Parkinson's. • Argues against aggressive amplitude normalisation.
Seifert et al. [4]	• Rhythm and pace parameters are reliable. • Joint velocity and event timing depend on geometry.	• Interpretation restricted to trustworthy components. • Acquisition geometry reported explicitly.
Papanastasiou et al. [9]; Ni and Huang [10]	• Micro-Doppler identifies individuals at over 93 %.	• Gait is a fingerprint. • Leave-one-subject-out validation non-negotiable.
Seyfioğlu et al. [16]; Park et al. [17]	• Pretrained networks transfer well to new radar tasks.	• Motivates the ResNet-18 transfer branch. • A frozen backbone suits a 58-subject cohort.
Erol et al. [18]	• ACGAN synthesises realistic micro-Doppler. • Expands small training sets.	• Supports data augmentation. • Informs the augmentation recipe of Chapter 3.
Cai et al. [19]	• STRIDE: radar gait analysis for dementia-risk screening.	• Radar gait analysis generalises beyond Parkinson's.
Meral and Ozbilgin [20]	• PD diagnosis from gait signals with deep learning. • Force sensors, not radar.	• PD gait carries a learnable signature. • Radar remained unexplored for subject-independent PD classification.

2.8 Research gap and positioning of this thesis

The reviewed literature supports three conclusions.

1. **What radar measures best is not what identifies the disease.** Radar micro-Doppler is a validated, information-rich way to record gait, but the quantities it measures most reliably (rhythm and pace) are not the ones most specific to Parkinson's (variability and absolute velocity) [4, 5].
2. **Deep learning on micro-Doppler is mature, but not for this task.** It performs well on broad problems such as activity recognition, person identification and young-versus-elderly discrimination, yet it has not been applied to subject-independent Parkinson's classification [7, 9, 10, 6].
3. **Headline accuracies in this field are not trustworthy.** They are repeatedly undermined by limitations that recur from work to work: random hold-out splits that let the same person appear in training and test, shortcut learning on recording artifacts, cohorts of three to twenty-two subjects, an uncontrolled confound in **age** [7], and, for Parkinson's disease, no radar classifier at all.

Closing this gap is the purpose of this thesis, stated in Chapter 1, on the multi-node FMCW dataset of López-Delgado et al. [6]. Methodologically, the work adopts together the safeguards that the literature identifies as necessary but rarely combines:

- **Leave-one-subject-out cross-validation** with subject-level score aggregation, to eliminate identity leakage.
- **Fixed-length windowing**, to suppress a trial-duration confound.
- **Explicit null floors**: deliberately simple reference predictors, recording duration alone and window count alone, reported beside every model to guard against the confounds highlighted by [7].
- Subject-level **bootstrap confidence intervals**, for honest uncertainty reporting.
- **Gradient-based attention checks** (Grad-CAM), to verify that the network attends to the Doppler bands of the feet and the torso rather than to artifacts.

One safeguard the literature calls for could not be adopted: the dataset records no ages, so the age confound is not controlled and is carried as a limitation (Section 5.2). An uncontrolled confound can only make a classifier look better than it is, so it cannot rescue a result that already sits at the duration floor.

In doing so, the work aims to establish not merely whether Parkinson's gait is detectable from radar, but whether it is detectable *under validation conditions strong enough to be trusted*.

3. Materials and Methods

This chapter describes the material this thesis works with and the methods applied to it. Sections 3.1 to 3.4 establish **what the data physically is**, how each recording was verified and how a recording is reduced to a short list of numbers. Section 3.5 defines the statistical tools used to compare the two groups. The remaining sections cover the models and the validation protocol.

3.1 Dataset

3.1.1 Provenance

Every measurement analysed in this thesis originates from a single acquisition campaign carried out at the Universidad Politécnica de Madrid and published by López-Delgado *et al.* [6]. That study validates a network of 24-GHz FMCW radar nodes against an optical motion-capture reference and establishes that the network measures temporal and spatial gait parameters accurately, including for motorically impaired participants. It stops at **measuring gait parameters** and contains **no disease classifier**, which is precisely the gap the present work addresses.

The acquisition environment is reproduced in Figure 3.1. Two tripods, each carrying one trunk-height node and one ankle-height node, face one another at the ends of a walking platform, and an eight-camera Vicon system provides the ground truth against which the radar was validated [6].

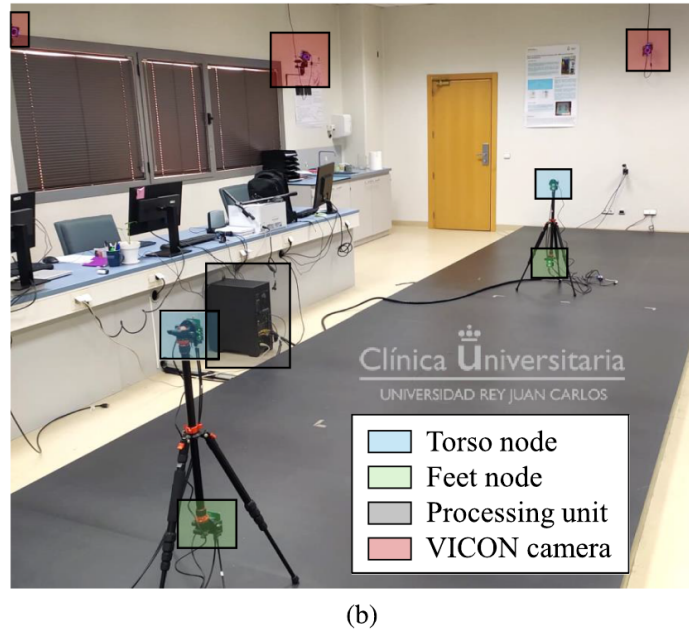
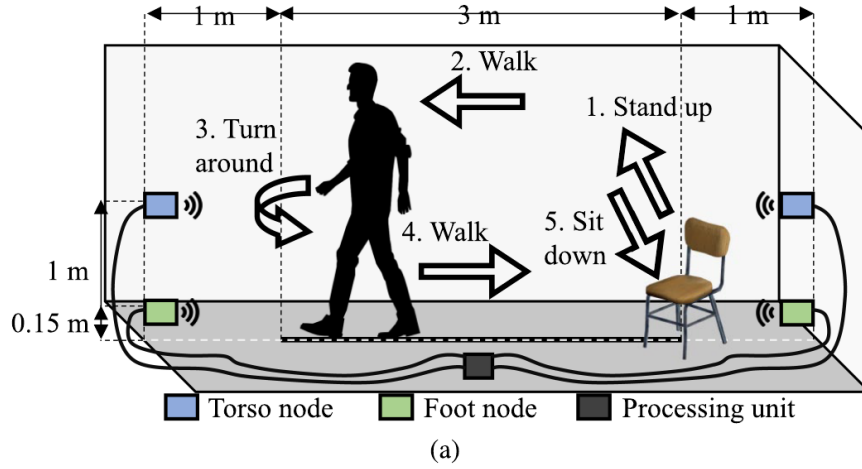


Figure 3.1: The acquisition environment as published. Above, the corridor layout with the five phases of the Timed-Up-and-Go test numbered in order and the chair at the right-hand end. Below, a photograph of the same room. Blue marks a torso node at 1 m, green a foot node at 0.15 m, dark grey the processing unit, and pink a Vicon camera. Reproduced from [6].

3.1.2 Acquisition protocol

Each subject performed the **Timed-Up-and-Go** (TUG) test, a standard clinical mobility assessment: stand up from a chair, walk 3 m in a straight line, turn around, walk back and sit down [6]. Two variants were recorded, three times each:

- Variant **test1**, the full protocol **including the chair**, so it contains a sit-to-stand transition and a stand-to-sit transition;
- Variant **test2**, the same walk and turn **without the chair**.

This single protocol difference proves to be the most consequential property of the dataset, for reasons developed in Section 4.1.2. Rising from a chair is among the movements most

affected by Parkinson’s disease, so any group difference in elapsed time is expected to concentrate in test1.

3.1.3 Cohort

The dataset comprises **58 subjects** and **348 recordings**, split as shown in Table 3.1. The intended design is **six recordings per subject**: two variants by three repetitions. Fifty-six of the 58 subjects match it exactly. Two PD subjects deviate: one contributed an extra test1 repetition (seven recordings, which is why test1 holds 76 PD recordings rather than 75), and one is missing a test2 repetition (five recordings, which is why test2 holds 74 rather than 75). Both cases are examined in the quality control of Section 3.1.5 and kept.

Table 3.1: Composition of the dataset.

Group	Subjects	test1 recordings	test2 recordings
Healthy control	33	99	99
Parkinson’s disease	25	76	74
Total	58	348 recordings	

A cohort of 58 subjects is the **defining constraint of this project**. It shapes every subsequent decision: compact models rather than large ones, reuse of networks pretrained elsewhere rather than training from scratch, and a validation protocol that holds out one subject at a time rather than relying on a single fixed test split.

The dataset provides the diagnostic label, the test variant and the repetition index. **Age, sex, disease severity and medication state are not recorded**. Age is the most important missing item: Parkinson’s disease is a disease of later life, and ageing alone shortens step length and slows movement [2], so a group difference admits two explanations that the available metadata cannot separate. Every result reported in this thesis is therefore a difference between **these two groups as recruited**, and this limitation is revisited in the discussion.

3.1.4 Contents of one recording

Each recording is stored as a MATLAB v5 file holding four spectrogram arrays and two axis vectors. Only two of the arrays are used in the main analysis:

- Channel **ce_foot**, the contrast-enhanced **foot-level** spectrogram;
- Channel **ce_torso**, the contrast-enhanced **trunk-level** spectrogram;
- A **Doppler axis** of 320 values spanning ± 800 Hz and a **time axis** whose length varies with the duration of the walk.

Each stored channel **combines the two same-height nodes** at the opposite ends of the corridor: at every time frame the frame from the node with the higher signal-to-noise ratio is kept, with the far node’s Doppler axis flipped first so that both describe motion

in the same direction [6]. The files also carry the four per-node arrays and the per-frame selection indices, which the main analysis does not use.

The arrays arrive already oriented as Doppler against time, so no transposition is required. Because the two enhanced spectrograms and the two axes are the only variables read in normal operation, loading one recording costs about 0.12 s and the full 106 GB dataset is never read in its entirety.

Contrast enhancement refers to a per-recording rescaling applied by the originating team before storage. Its purpose is straightforward: the trunk reflects far more energy than a foot, so on an unscaled image the trunk would saturate and the feet would be nearly invisible. The rescaling brings both into view simultaneously. The exact operation is not published, so the enhanced arrays are treated as the given input rather than as a reproducible step.

3.1.5 Quality control

All 348 recordings were verified before any analysis, so that a damaged file could not silently become a finding. A first stage checks structure without loading the large arrays: the expected variables are present, the four spectrogram arrays share one shape, the Doppler axis spans the expected range, and the duration is plausible. A second stage loads the two spectrograms and confirms that every value is finite and that the image carries genuine contrast.

All 348 recordings pass every check, so nothing was excluded and the modest results reported later **cannot be attributed to damaged inputs**. The two irregular subjects were examined rather than assumed: the extra recording is a genuine repeat attempt, not a duplicate, and the missing one is a clean deletion, so both subjects were kept.

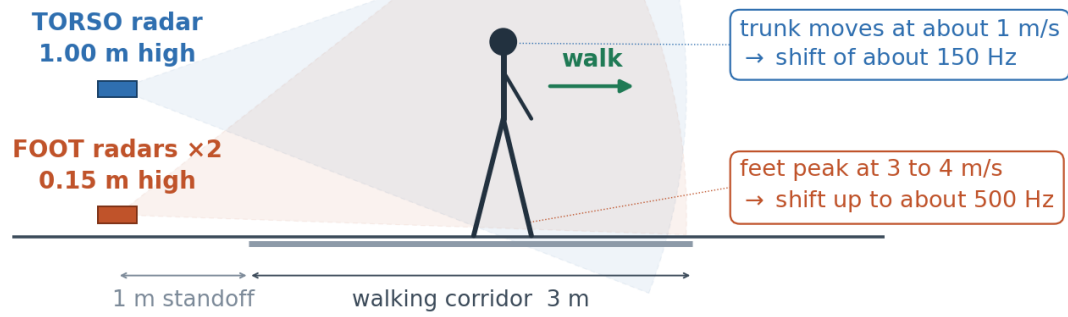
3.2 Radar System and Signal Model

3.2.1 From a transmitted chirp to a spectrogram

An FMCW radar transmits a **chirp**, a brief burst whose frequency sweeps upward, and compares the returning echo against the burst currently being transmitted. The difference between the two encodes both the range of the reflector and its radial velocity. Repeating this produces a continuous stream of measurements, and a short-time Fourier transform of that stream produces the images analysed here. Figure 3.2 summarises the geometry and the processing chain.

Acquisition geometry: two radars watching the same walk

four nodes on two tripods: at each end, one aimed at the trunk and one at the feet, 40° beamwidth



From radio wave to the two pictures we analyse



Steps 1 to 4 were run by the team that built the system. This thesis starts from the output of step 5.

Figure 3.2: Above, the acquisition geometry drawn from the published parameters. Below, the five steps that turn a transmitted chirp into the two arrays analysed in this thesis. Steps 1 to 4 were performed by the originating team; this work begins from the output of step 5.

Every parameter in Table 3.2 was fixed by the originating team rather than chosen for this thesis, so they are inherited constraints. They are reproduced unchanged from [6].

Table 3.2: Acquisition parameters and what each one controls.

Parameter	Value	What it controls
Sweep start frequency	23 GHz	The chirp sweeps upward from here, placing the system in the 24 GHz band; it sets the wavelength , about 1.3 cm, the conversion factor between frequency shift and velocity.
Chirp bandwidth	1.4 GHz	Range resolution : how finely two reflectors at different distances are separated.
Chirp duration	625 μ s	One measurement every 625 μ s is 1600 per second , which caps the measurable Doppler shift at half that rate and gives the ± 800 Hz span.
Clutter filter	10 Hz high-pass	Removes reflections from stationary objects such as walls and furniture.
Analysis window	50 ms Hann	The time against frequency trade-off ; the Hann taper separates components roughly 30 Hz apart (20 Hz for an ideal window).
Doppler bins	320 bins, ± 800 Hz	One stored value every 5 Hz.
Radar nodes	4	Two tripods, each holding one foot-level and one trunk-level node; the two same-height nodes are merged into the two stored channels .
Corridor / standoff	3 m / 1 m	A short walk, so recordings are brief: median 8.91 s.
Beamwidth	40°	The field of view, which sets when a subject enters and leaves the measurement.

3.2.2 Doppler shift and velocity

A moving reflector shifts the frequency of the wave that returns from it. For a monostatic radar the wave traverses the path twice, so the shift carries a factor of two:

$$f_D = \frac{2v}{\lambda}, \quad \text{equivalently} \quad v = \frac{f_D \lambda}{2}, \quad (3.1)$$

where v is the radial velocity of the reflector and $\lambda = c/f_0 \approx 1.3$ cm, computed at the 23 GHz sweep start (taking the centre of the swept band instead shifts every conversion by about 3%). A trunk advancing at 1.0 m/s produces a shift of approximately 153 Hz. Equation (3.1) is fixed by the hardware, so **the vertical axis of every spectrogram can be read as a radial-velocity axis**.

3.2.3 What a single spectrogram column represents

At any instant the radar receives reflections from **every moving part of the body at once**, and the Fourier transform separates them by velocity. A single spectrogram column is therefore a **distribution of the velocities present at that moment**, not the walking speed of the subject: brightness at height f means that much of the reflected energy came from body parts moving at the corresponding velocity.

3.2.4 The two analysis bands

Two frequency bands are used throughout. Their boundaries are inherited from the source publication, which places them from **established walking velocities** rather than fitting them to any particular dataset [6]:

- A **torso band** below **200 Hz**. A trunk at roughly 1 m/s produces about 153 Hz by Equation (3.1), and the boundary allows headroom for faster walkers.
- A **foot band** from **200 to 500 Hz**. A foot at peak swing reaches roughly 3.3 m/s, which is about 506 Hz.

The data supports the split: measured on a sample of this dataset, the torso channel concentrates its energy low and the foot channel sits higher, as expected. One caveat is kept in mind: about 21 % of the foot-channel energy lies above the 500 Hz boundary, so the bands **under-report the fastest foot motion**; widening them is left for future work.

3.2.5 Structure visible in a single recording

Three properties are directly legible from a spectrogram, and Figure 3.3 shows all three.

1. **The turn divides every recording in two.** Energy sits below zero for the first half of a recording and above zero for the second. A negative shift means motion **away** from the node and a positive shift means motion **towards** it, so the sign change marks the instant the subject turned around. This is clinically relevant because turning is the slowest and least stable phase of the walk, and impaired turning is characteristic of Parkinson's disease.
2. **The repeating arches are individual steps.** In the foot channel each arch is one foot swing, accelerating from rest, peaking mid-swing and decelerating to **heel strike** (foot touching ground). The valleys between arches are the intervals during which the foot is flat on the ground at zero velocity, so steps can be counted directly from the image.
3. **The two channels capture different physics.** The foot channel is spiky and periodic and dominated by 200 to 500 Hz swings; the torso channel is a single smooth band near 0 to 200 Hz.

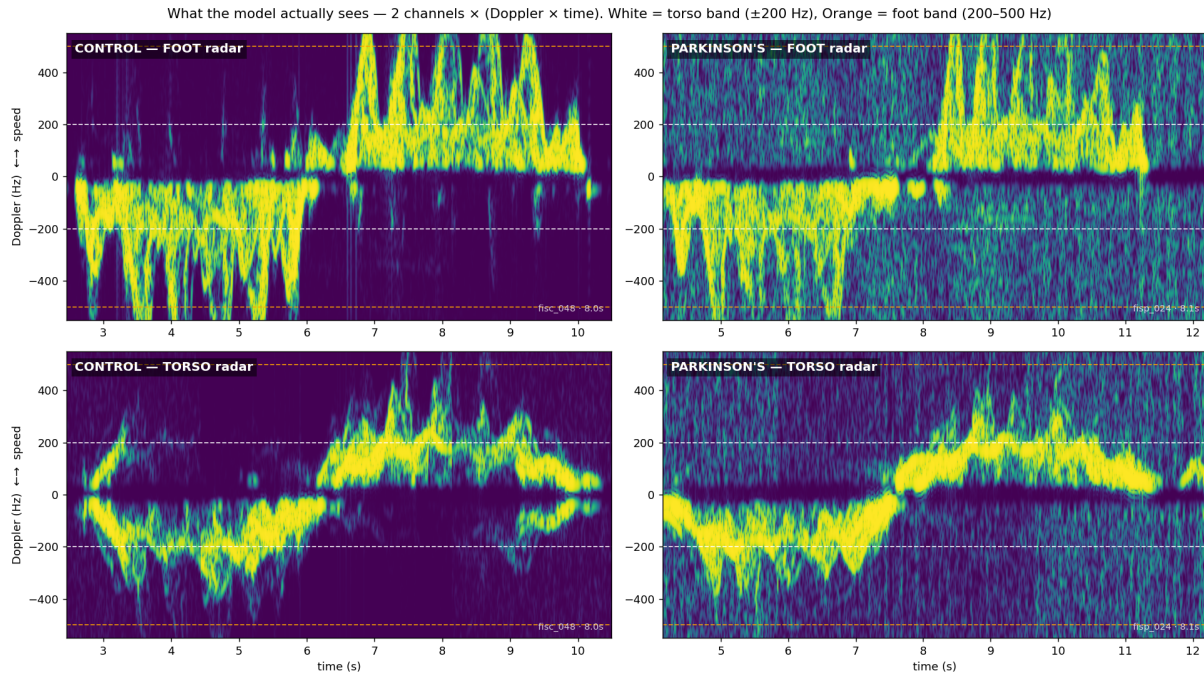


Figure 3.3: One control subject and one PD subject, each seen simultaneously by both nodes. The two recordings are matched for duration (8.0 s against 8.1 s) so that elapsed time is not the visible difference. White dashed lines mark the torso band at ± 200 Hz and orange the foot band from 200 to 500 Hz. The two left panels are the same recording of the same subject viewed by the two nodes, and likewise for the two right panels.

3.2.6 Coupling between the two channels

In healthy walking the two channels are **locked to each other in time**, and the clearest expression of this is the heel strike. The source publication marks that instant independently in each channel [6]: it appears as a **local minimum of the foot channel's energy**, because a foot resting on the ground is momentarily stationary, and as a **local peak of the trunk's speed** read from the torso channel, because the trunk is at its fastest as weight transfers. In healthy walkers the two markers coincide stride after stride. Figure 3.4 draws both markers directly on the signature of a healthy recording of this dataset: the upper contour of the foot channel, which dips to a valley each time the foot is planted, and the trunk-speed trace on the torso band, which peaks at those same instants, stride after stride.

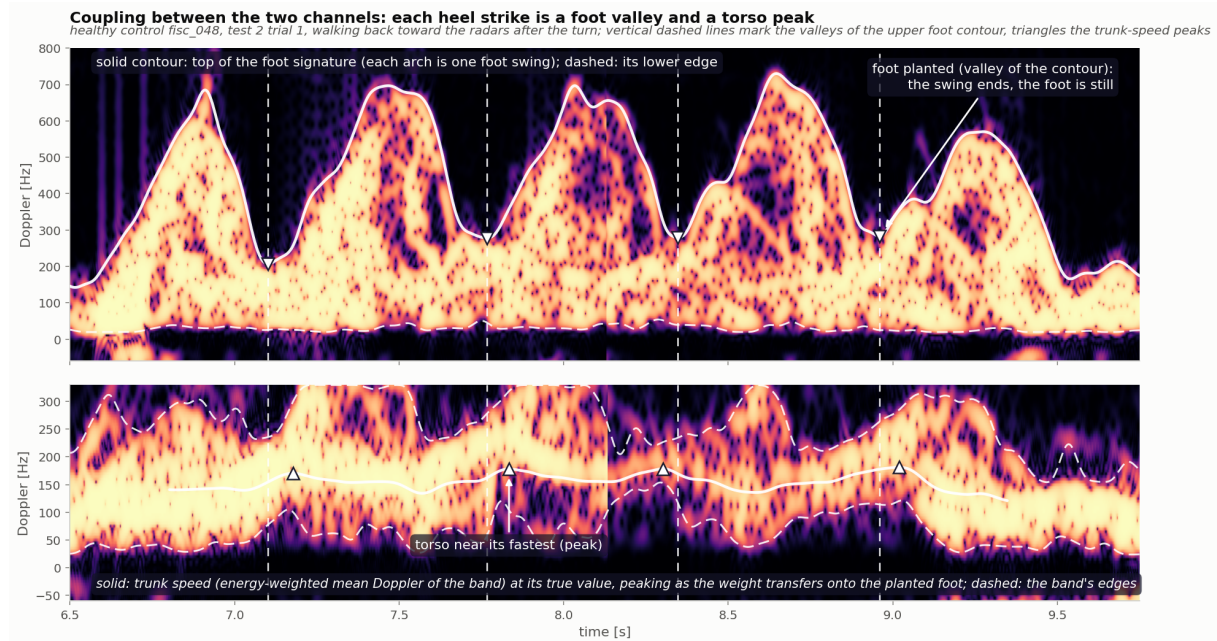


Figure 3.4: The coupling made visible on a healthy walker. In the upper panel the solid white contour follows the top of the foot signature and the dashed one its lower edge; the top contour dips to a valley each time the foot is planted (vertical dashed lines). In the lower panel the dashed contours outline the torso band and the solid white trace is the trunk's speed, drawn at its true Doppler value on the band; it is near a local peak at those same instants, so the two channels keep time with each other.

The same publication reports the observation that motivates the modelling choices in this thesis: that agreement is **lost in participants with Parkinson's disease**, so the trunk no longer tracks each heel strike reliably [6]. The information therefore resides not in either channel alone but in **whether the foot's rest and the trunk's speed keep time with each other**. A measurement taken from one channel in isolation cannot express that relationship, so any method intended to capture it must observe both channels simultaneously.

3.3 Preprocessing

Part of this chapter feeds the spectrogram itself to neural networks, and that raises two problems: a network needs inputs of one fixed size, and 348 recordings are too few to train one. A preprocessing pipeline addresses both, and Figure 3.5 shows the whole path.

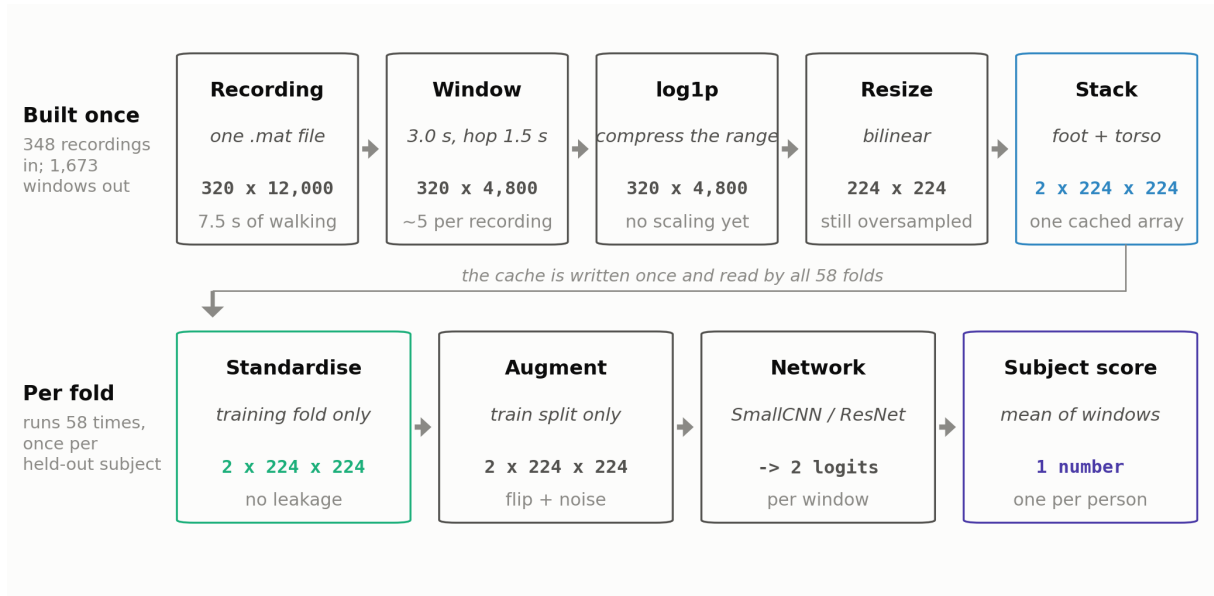


Figure 3.5: The deep-learning path from a stored recording to one score per person. The upper row runs once and writes a cache to disk; the lower row runs inside every fold of the validation loop, so that no operation performed there can see the held-out subject. The classical baseline of Section 3.6 follows its own flow and never sees windows.

Two phases of the figure matter for validity. The upper row runs once and writes a cache to disk. The lower row runs inside every round of the validation loop and applies only the operations that must not see the held-out subject. Throughout the rest of this chapter one such round is called a **fold**: one subject is set aside for testing, every data-dependent operation is performed on the remaining subjects only, and the round is repeated with each subject in turn (Section 3.8).

Table 3.3 explains each stage of Figure 3.5 and traces the shape of the data through the pipeline: a recording enters as $320 \times 12,000$ numbers and leaves as one number between 0 and 1.

Table 3.3: The stages of Figure 3.5: what each one does and the shape it leaves behind.

Stage	What it does	Shape after it
Windowing	Cuts the recording into 3.0 s pieces, one starting every 1.5 s (Section 3.3.1).	$320 \times 4,800$
Amplitude compression	Evens the brightness gap between the strong torso and the faint feet by storing $\log(1 + x)$ instead of x .	unchanged
Resizing	Shrinks each window to the input grid of the pretrained network.	224×224
Channel stacking	Joins the foot and torso windows into one two-channel array.	$2 \times 224 \times 224$
Standardisation	Centres the values, using statistics from the training subjects of the fold only.	unchanged
Augmentation	Training windows only: a possible flip along the Doppler axis plus a little noise; test windows pass unchanged.	unchanged
Network	Scores one window: a control score and a PD score.	2
Window probability	Converts the two scores into one probability of PD.	1
Subject score	Averages one person's window probabilities: the reported score.	1

Several design choices in this section admit a reasonable alternative. Each such alternative is evaluated by an **ablation**: the removal or replacement of one single component of the pipeline, with everything else held fixed, in order to measure how that component affects the overall result. The choices are thereby measured rather than asserted; the ablation results are reported in Chapter 4. One variant is computed from the raw STFT magnitudes rather than the enhanced arrays of Section 3.1, so the representation itself is also tested.

3.3.1 Windowing

Each recording is cut into windows of **3.0 s**, a new window starting every **1.5 s**, so that consecutive windows share half their content. The length is set by the gait itself. One stride takes approximately 1.0 to 1.2 s, and stride-to-stride regularity, one of the clinically motivated quantities of Section 3.4, requires at least two complete strides in view; a 3.0 s window always contains two to three. The length was fixed in advance from this reasoning rather than tuned; treating it as a hyperparameter and selecting it with the AUC as figure of merit is left for future work (Section 5.4). The overlap is deliberate, for two reasons:

- Advancing by a full window length would halve the number of training examples drawn from an already small dataset, and

- A stride falling across a cut would be split in two and seen whole by no window, whereas with a half-window overlap every moment of the walk appears complete in at least one window.

Figure 3.6 shows the cuts drawn onto a real recording.

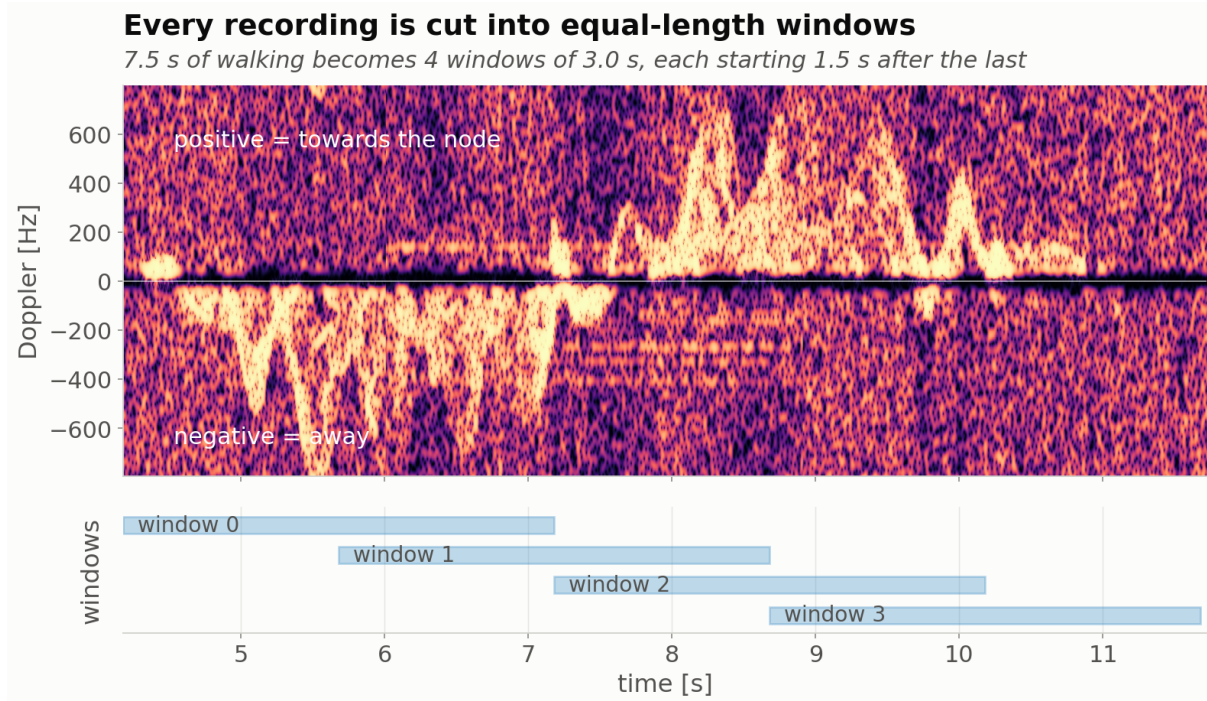


Figure 3.6: Windowing of one real recording. Above, the cut points drawn onto the spectrogram; below, the resulting windows as bars, making the half-window overlap visible: each window starts 1.5s after the previous one and lasts 3.0s.

Recordings differ in duration, so they yield between 2 and 11 windows each (4.8 on average), and the dataset of 348 recordings becomes **1,673 windows**. A finer setting of 2.0s windows every 0.5s was considered and rejected: neighbouring windows would then share 75% of their content, so the additional windows are near-copies rather than new information.

3.3.2 Fold-local standardisation

Networks train poorly on inputs that are not centred near zero, so the cached values are additionally **standardised** to mean 0 and standard deviation 1. The statistics are computed from **the training subjects of each fold only** and applied at loading time; the cache itself holds unstandardised values. Computing the statistics over the whole dataset would let the held-out subject influence the scale against which its own input is later measured, a small leak, but the argument of this thesis requires there to be none.

A common alternative standardises each window against its own mean and spread. It is not used here, because it deletes the window's absolute brightness, and that single deletion would remove two different things at once:

- A possible **signal**: brightness reflects how strongly the moving body scatters, which bradykinesia plausibly changes;

- A known **nuisance**: overall brightness also varies between recordings for reasons unrelated to gait, such as the per-recording contrast enhancement described in Section 3.1.

Because it cannot be known in advance which of the two dominates, the default pipeline keeps absolute brightness and the per-window variant is evaluated as an ablation, with the comparison reported in Chapter 4.

3.3.3 Resampling and channel stacking

Each window is resized independently from $320 \times 4,800$ to 224×224 by **anti-aliased bilinear interpolation**; the target size is not tuned but fixed by the input layer of the pretrained network of Section 3.7. Both axes remain oversampled after resizing, so the operation discards duplication rather than detail. The two resized channels, foot-aimed and torso-aimed, are then stacked into a single $2 \times 224 \times 224$ array, in the same way a colour photograph stacks its three colour layers. Figure 3.7 shows one cached window exactly as the network receives it: the foot channel is spiky and reaches high speeds, the torso channel is a smooth band near zero, and the disease-relevant coupling between the two (Section 3.2) exists only when both are present.

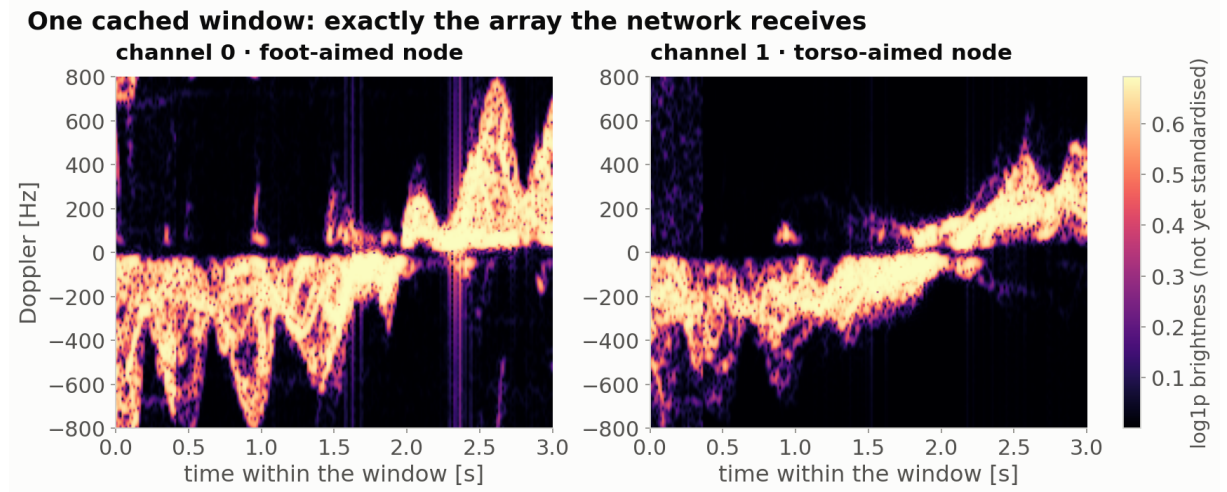


Figure 3.7: One cached window after compression, resizing and stacking: channel 0 is the foot-aimed node and channel 1 the torso-aimed node, recorded during the same three seconds of the same walk.

Each window is stored as one file, and an **index table** records one row per window: the subject, their diagnosis, the source recording, the position within it, and the file location. Training operates exclusively on this index, so selecting any group of subjects reduces to filtering the table.

3.3.4 Window weighting

Fixing the window size removes recording length from the *shape* of the input but not from the *number* of inputs: longer recordings yield more windows. PD subjects contribute 30.9 windows on average against 27.3 for controls, and window count alone separates the groups at a subject-level AUC of 0.62. The network never sees the count itself, but without a

correction the subjects with more windows dominate the training loss, and the network is pushed to tune itself to whatever the longer recordings have in common, which is exactly the group difference in duration. The correction is a single multiplication: each window's contribution to the training loss is scaled by the inverse of its subject's window count, so that every subject carries exactly the same total influence regardless of how long their recordings are. Figure 3.8 shows the shares before and after the correction.

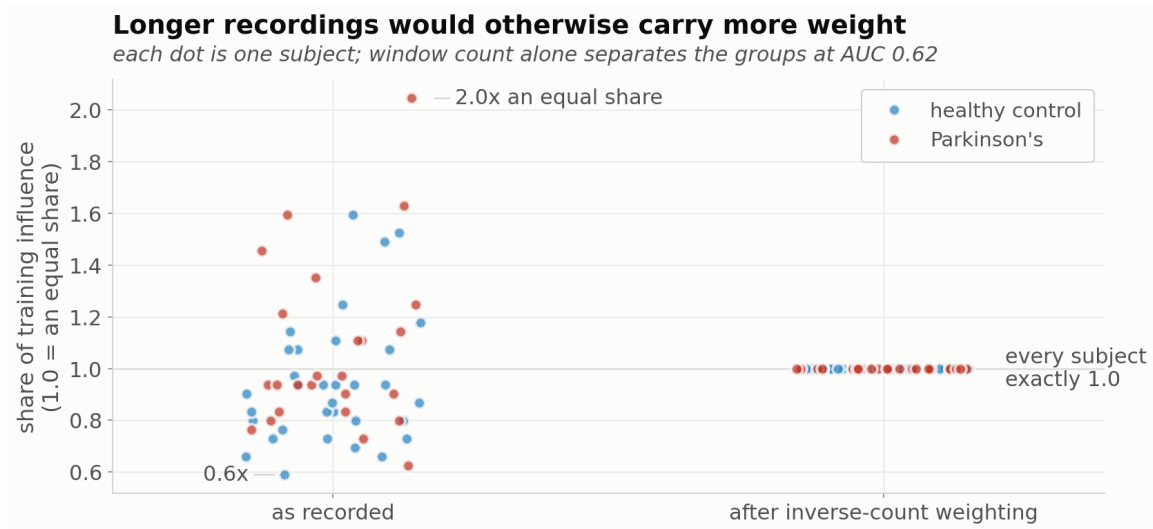


Figure 3.8: Share of training influence per subject, as recorded (left) and after inverse-count weighting (right). Each dot is one subject; without the correction the shares span more than a factor of three, driven by recording length.

3.3.5 Direction-of-travel labelling

The final stage attaches to every window the fraction of its foot-channel energy located at positive Doppler frequencies, measured after subtracting the recording's own background level. Values near 0 or 1 identify windows lying wholly inside the outward or the return pass; values near 0.5 identify windows containing the turn, where energy is present in both directions at once. The label makes a targeted experiment possible. The source publication computes its gait parameters only after discarding the turning, standing and sitting stages [6], whereas this pipeline keeps everything, and the exploratory analysis of Chapter 4 shows that the additional time PD subjects take is concentrated precisely in those discarded stages. Excluding all windows within 0.05 of the midpoint retains 54 % of the dataset, at the cost of four subjects losing every window, and is evaluated as an ablation that attacks this confound at its source. Figure 3.9 shows the distribution of the label and the retention cost of excluding the turn.

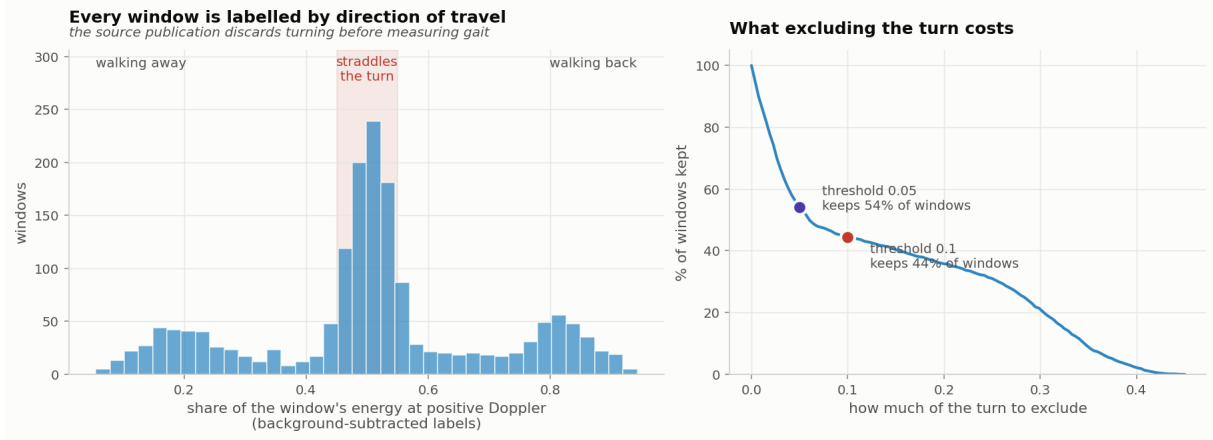


Figure 3.9: Direction-of-travel labelling. Left: share of each window's foot-channel energy at positive Doppler; windows near 0 or 1 lie inside a single walking pass, windows near 0.5 contain the turn. Right: fraction of the dataset retained as a function of the exclusion threshold.

3.4 Handcrafted Features

A micro-Doppler signature is a grid of roughly $320 \times 12,000$ values. Statistical comparison between groups requires instead a **short, fixed-length description of each recording**, so every recording is summarised by the same procedure: a set of descriptive statistics is computed over the whole of the foot spectrogram, the identical calculation is repeated on the torso spectrogram, and two cross-channel ratios are appended. The result is **one row of 32 numbers per recording**, giving a table of 348 rows.

Of those 32, **25 are carried into the analysis**: 17 assigned to the length-independent set and 8 to the length-dependent set (Table 3.4). The remaining 7 are redundant variants, namely the four per-channel band means, the two alternative normalisations of the foot-to-torso ratio, and a cross-channel ratio of the two channel means, which duplicate information already present and are excluded to avoid inflating the number of statistical tests.

3.4.1 Behaviour with respect to recording length

The features are divided into two classes, and the distinction governs every later analysis. Table 3.4 lists them.

Table 3.4: Handcrafted features and their behaviour with recording length.

Feature	Physical interpretation	Grows with length
Mean brightness	Overall reflected energy density	no
Peak brightness	The single strongest instant of motion	no
Brightness dispersion	How much the signal varies	no
Spectral centroid	Mean velocity of the body , the balance point of the energy on the Doppler axis	no
Spectral bandwidth	Range of velocities present: uniform or varied motion	no
Spectral entropy	How evenly energy is spread across velocities	no
Foot-to-torso ratio	Share of the motion that is fast (feet) rather than slow (trunk)	no
Rhythmicity	Regularity of stride timing , measured relative to the mean	no
Band and total energy sums	Energy accumulated over the whole recording	yes
Absolute envelope variability	Unscaled variation of energy over time	yes

3.4.2 Features chosen for clinical reasons

Three features were selected because each should respond to a specific, clinically established motor sign, and because the **direction** of the expected response can be stated in advance. The signs themselves are taken from the clinical literature rather than observed here: **bradykinesia** is the one motor sign the international diagnostic criteria require before a diagnosis of Parkinson's disease can be made [1], while **shorter steps with reduced foot clearance** and **increased stride-to-stride variability** are the gait changes most consistently reported in reviews of the condition [2, 3]. Table 3.5 maps each sign onto a feature.

Table 3.5: Clinically motivated features and their predicted direction.

Motor sign	Effect on the walk	Predicted effect on the feature
Bradykinesia (slowness of movement) [1]	Every body part moves more slowly, so all Doppler shifts are smaller.	The spectral centroid falls , because energy sits lower on the velocity axis.
Reduced foot clearance (shuffling, shorter steps) [2]	The foot travels less far and less quickly, so less energy reaches the fast band.	The foot-to-torso ratio falls , because the fast share of the motion shrinks.
Gait variability (irregular stride to stride) [3]	Steps become less evenly timed, so energy over time is less regular.	Rhythmicity rises , because variation relative to the mean increases.

Committing to these directions in advance converts each feature into a **testable prediction** rather than an observation to be rationalised afterwards. If the centroid rises where the disease predicts a fall, the difference cannot be the sign the feature was chosen to detect, and something else is driving it.

3.5 Statistical Analysis

The quantities defined in this section, the probability value p , the effect size δ and the corrected probability value, appear in nearly every figure and table of Chapter 4; defining them once here keeps those captions short. All group comparisons are performed on **one feature at a time**. For a given feature, its value is read from every recording, producing $n_1 = 150$ values labelled PD and $n_2 = 198$ labelled control, and the three quantities below are computed from those two lists. All three derive from the same operation: comparing every PD value against every control value and counting how often the PD value is larger.

One caution applies to everything in this section. The tests treat the 348 recordings as independent, but each subject contributes about six, and Chapter 4 measures directly that recordings of the same person resemble each other far more than recordings of different people. There are therefore fewer truly independent observations than the counts suggest, and the probability values are optimistic. This is acceptable because of the role they play: they describe the dataset, and no conclusion of this thesis rests on them. Every claim that matters is established by the subject-level validation protocol of Section 3.8 and by confidence intervals that resample subjects, never recordings (Section 3.10).

3.5.1 Rank-based comparison

The **Mann-Whitney U** statistic [21] is that count,

$$U = \#\{(x_i, y_j) : x_i > y_j\} + \frac{1}{2} \#\{(x_i, y_j) : x_i = y_j\}, \quad (3.2)$$

where x ranges over the PD values and y over the control values. Under the null hypothesis that the two groups are interchangeable, U is centred on $n_1n_2/2$, and the **probability value** p is the probability, under that hypothesis, of a value at least as far from the centre as the one observed, u :

$$p = \Pr_{H_0}(|U - \frac{n_1n_2}{2}| \geq |u - \frac{n_1n_2}{2}|). \quad (3.3)$$

A rank-based test was preferred to a t -test because several of the features are strongly skewed, and a mean is easily displaced by a small number of extreme recordings whereas a rank can only ever win its own pairwise comparisons.

3.5.2 Effect size

The probability value reports whether a difference is likely to be real, not whether it is large. **Cliff's delta** [22] reports the magnitude on a scale independent of sample size:

$$\delta = \frac{\#\{x_i > y_j\} - \#\{x_i < y_j\}}{n_1n_2} = \frac{2U}{n_1n_2} - 1. \quad (3.4)$$

Here $\delta = 0$ means the two groups are interchangeable and $\delta = 1$ means every PD value exceeds every control value. The magnitude labels used throughout follow the conventional thresholds of 0.147, 0.33 and 0.474 for small, medium and large [23]. Both quantities are reported for every comparison, because a sufficiently large dataset renders a negligible difference statistically convincing.

3.5.3 Correction for multiple comparisons

Testing 25 features in succession would produce approximately one apparently convincing result by chance alone. Probability values are therefore corrected with the **Benjamini-Hochberg** procedure [24]. With m tests and $p_{(1)} \leq \dots \leq p_{(m)}$ the sorted probability values, the corrected value at rank i is

$$p_{(i)}^{\text{fdr}} = \min_{j \geq i} \min\left(1, \frac{m}{j} p_{(j)}\right). \quad (3.5)$$

The procedure controls the expected proportion of false positives among the reported discoveries. It was preferred to the stricter familywise alternative, which is valid under any dependence but so conservative on 25 strongly inter-correlated features that genuine differences would be discarded; the Benjamini-Hochberg guarantee holds under the positive dependence typical of such features.

3.5.4 Subject-level similarity

To quantify how identifiable an individual is from a single walk, each recording is represented by its **17 duration-independent features**, each standardised to zero mean and unit variance so that no feature dominates by virtue of its numerical scale. A recording is then a point in $\mathbb{R}^{17}$ and the dissimilarity between two recordings is the Euclidean distance

$$d(a, b) = \left(\sum_{k=1}^{17} (a_k - b_k)^2 \right)^{1/2}, \quad (3.6)$$

expressed in units of one standard deviation. Restricting the calculation to duration-independent features is deliberate: were the summed features included, a subject who consistently walks slowly would appear identifiable through elapsed time rather than through the shape of their gait. For each subject two quantities are formed, the **within-subject** distance, averaged over all pairs of that subject's own recordings, and the **between-subject** distance, averaged from that subject's centroid to ten recordings of other subjects drawn at random with a fixed seed.

3.6 Classical Baselines

The deep models are the goal of this thesis; the classical classifiers in this section are the **yardstick** against which they are read. Before any network is trained, three ordinary statistical classifiers are fitted to the 25 handcrafted features of Section 3.4, and whatever they score becomes the bar a network must clearly exceed before anything can be claimed for it.

One classical prediction is produced in five stages, summarised in Figure 3.10 and repeated once for each subject in turn:

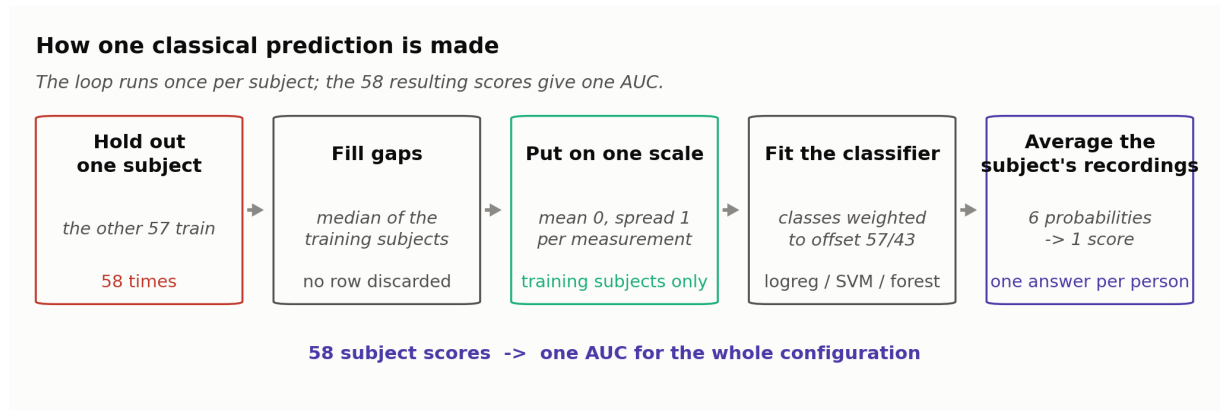


Figure 3.10: One fold of the classical baseline. The loop runs once per subject; the 58 resulting per-subject probabilities give one AUC per classifier and feature set.

1. **Hold out one subject.** All of that person's recordings are set aside and touch nothing that follows.
2. **Fill gaps.** A guard rather than a repair: any feature undefined for some recording would be replaced by the median of that feature across the training subjects. On this dataset the guard never fires, since all 348 recordings yield all 25 features; the rule is stated because a protocol must fix in advance what happens when a value is missing.
3. **Standardise.** The features span wildly different units, so each is rescaled to mean 0 and spread 1 using the training subjects only.

4. **Fit with class weighting.** The cohort is 57 % control and 43 % PD, so a classifier that always answers control would already appear 57 % correct. Errors on each class are therefore scaled by the inverse of that class's share, giving the smaller PD group the same total influence on the fit as the larger control group.
5. **Score and average.** The classifier outputs a probability of PD, one per recording, and the held-out subject's recordings are averaged into **one probability per person**. Only after all 58 folds are those 58 probabilities compared with the true diagnoses.

Averaging per subject yields the answer a clinician would want, one per person, and makes the score insensitive to how many recordings a subject happens to have, the classical counterpart of the window weighting of Section 3.3.

Three classifiers are compared (Table 3.6), chosen not to find a winner but because they represent different kinds of pattern, so that their agreement or disagreement is itself informative: if only the first succeeds, the signal is a simple trend; if only the third, it is conditional; if all three fail together, the reasonable reading is that no signal of any of these shapes is present at this sample size.

Table 3.6: The three classical classifiers and their roles.

Model	Why it suits this problem
Logistic regression	Assigns every feature a weight, adds up the evidence and converts the total into a probability. The most restrictive of the three, therefore the hardest to fool: with 58 subjects a flexible model can fit noise, this one largely cannot, and its weights are directly readable.
Support vector machine	Places a curved boundary between the groups, as far as possible from the nearest subjects, capturing signals that appear only in combinations of features.
Random forest	Averages the votes of many decision trees grown on random subsets, capturing thresholds and conditional patterns, and ranks the features by usefulness as a by-product.

Each classifier is additionally run on five nested **feature sets**, summarised in Table 3.7. The sets differ in exactly one respect, how much of the recording-length confound they can contain, so reading the resulting grid across is a controlled experiment: a score that survives the removal of length-carrying features reflects gait, and a score that falls step by step with them reflects recording length.

Table 3.7: The five feature sets of the classical baseline.

Set	Size	Question it answers
<code>all_features</code>	25	Performance with no restraint; the number most published work would report.
<code>duration_invariant</code>	17	Performance once length is removed by definition : totals grow with duration, averages and ratios do not, and only the latter are kept.
<code>shape_clean</code>	10	Performance once length is removed in practice : each remaining feature is checked directly against measured duration, and those that still track it (a longer recording holds more standing and turning, which drags some averages) are dropped. The rule is fixed in advance: any feature whose rank correlation with duration reaches 0.3, computed once over all recordings, is out. The check uses duration only, never the diagnosis, and can only remove information from the classifier.
<code>confounded_only</code>	8	The totals alone; how much the confound alone contributes.
<code>duration_only</code>	1	Recording length and nothing else; the floor . A result not clearly above it demonstrates nothing about gait.

3.7 Deep Models

With the yardstick defined and the input pipeline built, what remains is the choice of networks, and that choice is dominated by one number: 58 subjects. The 1,673 cached windows overstate the usable evidence, since overlapping windows of one walk are strongly correlated, so any architecture with more free parameters than windows can in principle memorise the training set and learn nothing general. The two primary architectures are therefore kept small and are deliberately chosen to bracket the problem from opposite directions, while multi-million-parameter variants are run only as capacity checks; a third model adapts the approach the closest literature itself trusts most. Figure 3.11 places every option against the number of available windows.

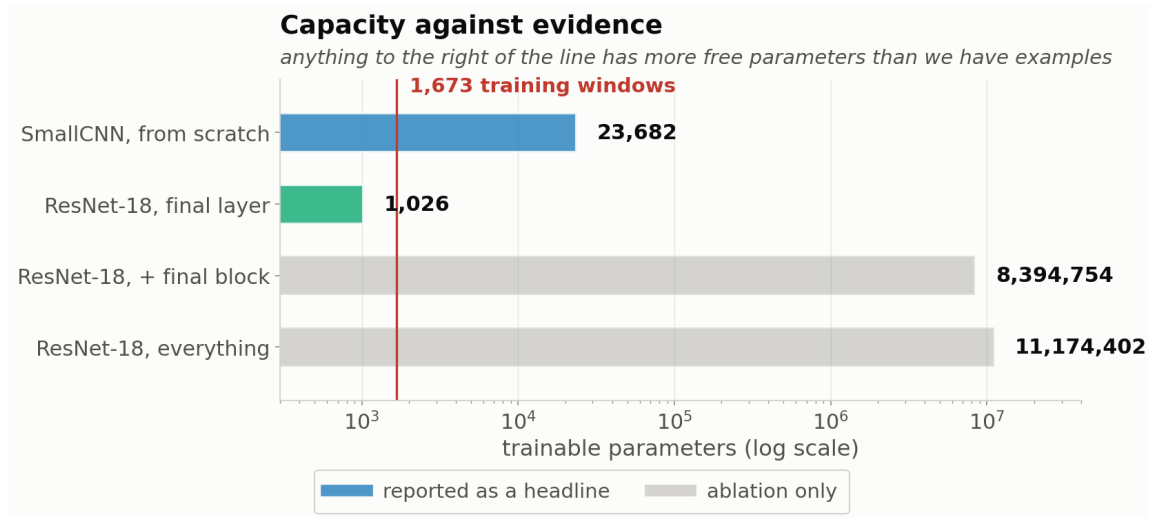


Figure 3.11: Trainable parameters of each candidate on a logarithmic scale, against the 1,673 available training windows. The multi-million-parameter variants can memorise the dataset outright and are run only as capacity checks.

3.7.1 SmallCNN, trained from scratch

The first network is purpose-built and small; Figure 3.12 draws it layer by layer.

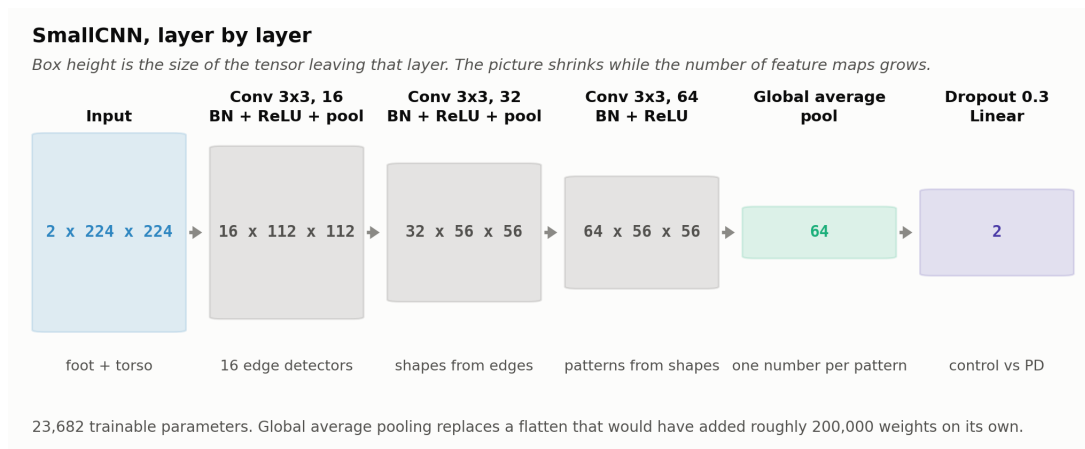


Figure 3.12: SmallCNN, layer by layer, with the shape of the tensor leaving each stage. The picture shrinks while the number of feature maps grows, and global average pooling replaces the flatten that would otherwise dominate the parameter count.

Three convolutional blocks of 16, 32 and 64 filters (3×3 , each followed by batch normalisation, rectification and, for the first two, spatial halving) feed a global average pooling layer, dropout of 0.3, and a linear layer to two class scores, for **23,682 trainable parameters** in total. Convolution recognises a local pattern wherever it occurs, which suits gait because a stride can begin anywhere in a window. The consequential choice is **global average pooling**: flattening the final feature maps into the class scores would require roughly 400,000 parameters in that single layer, seventeen times the rest of the network, whereas averaging removes them and forces each filter to carry meaning regardless of where in the window it fired.

3.7.2 ResNet-18 as a linear probe

The second network brings knowledge from outside the dataset. **Transfer learning** from ImageNet-pretrained networks is the best-performing route on small radar datasets [17, 10], and it is used here in its most conservative form. Two adaptations connect the pretrained ResNet-18 to this data. First, its input layer expects three colour channels; the red, green and blue filters are averaged into one, and the average is applied to both radar channels, preserving the pretrained edge and texture detectors rather than discarding them. Second, the entire backbone of 11.2 million weights is **frozen** at its pretrained values, including the internal normalisation statistics, which are pinned so that they cannot drift during training; only the final linear layer trains, with **1,026 parameters**. This configuration, a **linear probe**, asks a precise question: whether the two groups are already separable in a rich, fixed image representation, without allowing the network to reshape that representation around 58 people. Figure 3.13 draws the network with both adaptations marked, and Table 3.8 gives the parameter budget of each retraining depth.

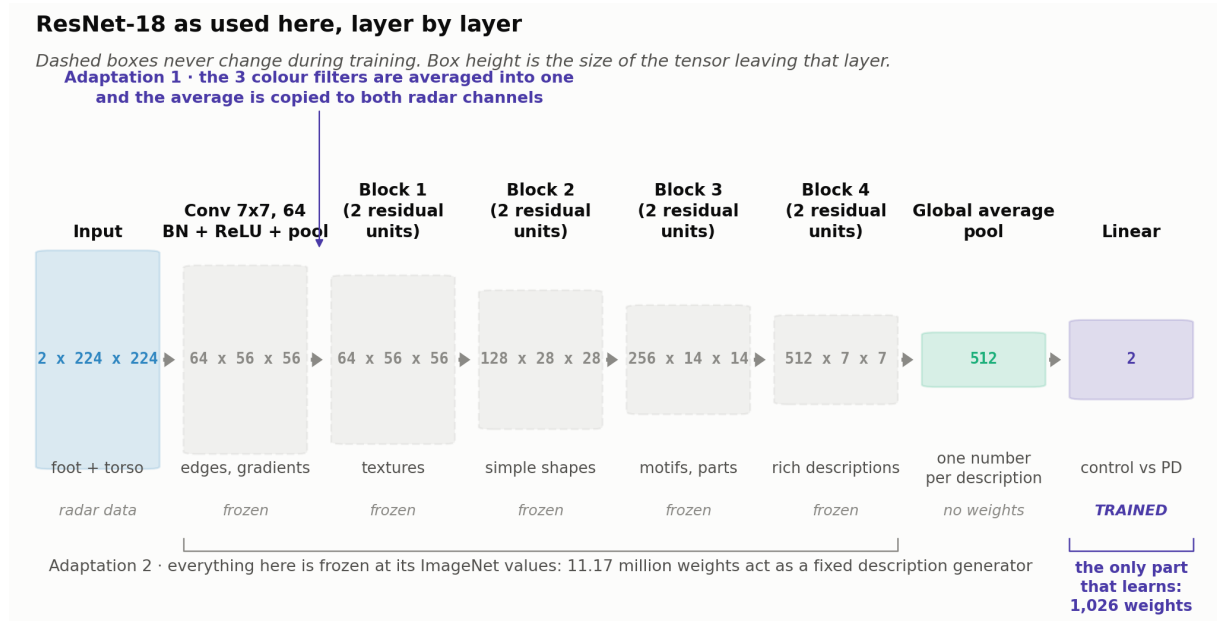


Figure 3.13: ResNet-18 as used here. Dashed stages are frozen at their ImageNet-pretrained values and act as a fixed description generator; only the final linear layer trains. The colour-filter averaging at the entrance is the only other modification.

Table 3.8: Retraining depth for the pretrained network.

Setting	Trainable	Role
Final layer only	1,026	Default: the linear probe.
Final block and layer	8,394,754	Capacity check; overfitting expected.
Everything	11,174,402	Not run; shown for scale.

Two findings of the source publication shape the experiments rather than the architecture [6] (Table 3.9).

Table 3.9: What the source publication contributes to the design.

What the paper establishes	What this design does with it
Foot-aimed nodes remain reliable on motorically impaired subjects; torso-aimed nodes lose reliability on exactly those subjects.	Predicts the foot channel carries more of the usable signal, turning the foot-only run of Chapter 4 into a test of a stated hypothesis rather than a sweep.
The trunk normally accelerates slightly at every heel strike; this trunk-to-step coupling weakens in Parkinson's.	The marker lives between the channels, so both are always presented to the networks together.

3.7.3 Envelope-LSTM

The third model adapts, to this dataset's combined channels, the design the closest related work preferred over its own higher-scoring alternative [7]. Instead of images it receives three velocity curves per window, each resampled to 128 steps: the upper envelope of the foot channel, tracking the fastest speed carrying meaningful energy, and the energy-weighted mean speeds of the foot and torso channels. The curves feed a bidirectional LSTM with 64 hidden units per direction, mean-pooled over time into a dropout-regularised linear layer, for 35,586 trainable parameters. By construction the per-recording image texture, and any artifact living in it, is stripped away before the model sees anything.

3.7.4 Training configuration

All deep models share the single recipe of Table 3.10. One choice is physically motivated and warrants explanation: the augmentation flips the **Doppler axis**, converting walking away from the node into the same walk towards it, a motion every subject genuinely performs twice per recording [15]. Flipping the time axis instead was rejected as physically wrong, since it plays a stride backwards, an accelerate-then-brake motion no walker produces.

Table 3.10: Shared training configuration of all deep models.

Choice	Value
Optimiser	Adam, learning rate 10^{-3} , weight decay 10^{-4}
Batch size	32 windows
Budget	at most 20 epochs, early stopping with patience 5
Class weighting	inverse class share, computed on effective (window-weighted) sample mass
Window weighting	inverse window count per subject (Section 3.3)
Augmentation	Doppler-axis flip ($p = 0.5$); Gaussian noise ($\sigma = 0.1$ on standardised data); random masking of one span of time columns and one of Doppler rows

3.8 Validation Protocol

Most published work in this field splits its recordings at random into training and test sets. That choice is easier, produces far higher numbers, and is not used here. Three findings, taken together, imply that a random split measures the wrong thing on this dataset:

1. **A walk identifies the walker.** The exploratory analysis of Chapter 4 finds each subject's recordings roughly 2.5 times more similar to each other than to anyone else's, for all 58 subjects without exception, and published systems identify individuals from radar micro-Doppler at accuracies above 93% [9, 10]. Identity is the strongest signal in this modality.
2. **A random split leaks it.** With about six recordings per person, a random split places some of a subject's recordings in training and others in test, so a model can score well by recognising the person and recalling the label seen in training, having learned nothing about the disease.
3. **The failure is documented, not hypothetical.** Of the sixteen works reviewed in Chapter 2, only two report subject-independent results [17, 15]. In the closest study, a spectrogram network reached 97.8% accuracy separating young from elderly walkers, and its own authors flagged the number as unreliable: the two groups had been recorded at different sites, and the network had learned each room's pattern of background noise, the ambient reflections filling the spectrogram behind the walker, rather than the gait [7].

After the protocol had been frozen and the campaign run, these findings were also put to a direct experimental test as a post-hoc diagnostic: the networks of Section 3.7 were retrained under the literature's own random splits, reproduced literature-grade scores, and collapsed to chance on subjects excluded from training entirely. That experiment opens the discussion of Chapter 5.

The deciding argument is simpler still. A validation protocol should match how the tool would be used, and a screening tool is applied to people it was never trained on. **Leave-one-subject-out** cross-validation enforces exactly that, 58 times over. Figure 3.14 shows the three groups of one fold with their widths to scale, and Table 3.11 states what each group decides. Each fold proceeds as follows:

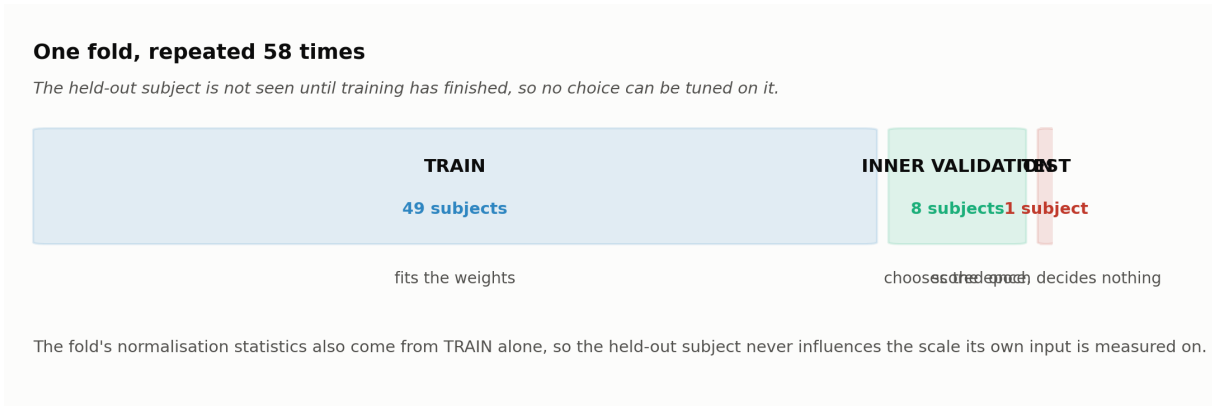


Figure 3.14: One fold of the nested protocol, widths drawn to scale. The held-out subject is scored exactly once and informs no decision; the fold is repeated 58 times, once per subject.

Table 3.11: The three groups of one fold and their roles.

Group	Size	What it decides
Train	49	The weights of the network, and the standardisation statistics.
Inner validation	8	Which training epoch to keep; nothing else.
Test	1	Nothing. It is scored once and reported.

1. One subject is set aside as the **test subject**, untouched;
2. The remaining 57 are split into 49 **training** subjects and an **inner validation** group of 8, drawn by a deterministic, stratified rule seeded by the held-out subject's identifier, so the split is reproducible and identical for every model; the group keeps close to the pool's control-to-PD proportion and always contains both classes, and its only job is to decide which training epoch to keep;
3. The standardisation statistics of Section 3.3 are computed from the 49 training subjects;
4. A fresh network is trained on the training subjects' windows and, after every epoch, scored on the inner validation group exactly as test subjects are scored, one averaged probability per subject;
5. The weights of the epoch with the highest subject-level AUC on the inner validation group are restored (when several epochs tie, the earliest wins), the test subject's windows are scored once, their probabilities averaged into one stored number, and the network is discarded;
6. The loop restarts from a blank slate with the next subject held out.

After 58 folds there are 58 numbers, one per subject, each produced by a model that never saw that person, and these are compared against the 58 diagnoses to give the single reported score. The classical baselines follow the same outer loop but omit the inner validation group: they fit in a single pass with fixed settings (the scikit-learn defaults,

version-pinned and recorded in the frozen configuration archived with the project), so there is no epoch to select and nothing for such a group to decide, giving 57 training subjects per fold. The inner group is the piece most published work omits, yet without it the only way to decide when to stop training is to watch the test subject, which is leakage. Leave-one-subject-out is preferred over grouped alternatives for three reasons:

- **Training data is the scarce resource**, and every fold uses the most of it that honesty allows;
- It yields a **personal score for every subject**, on which the uncertainty and confound analyses rest;
- It admits **exactly one possible set of folds**, leaving nothing to a fortunate draw.

The complete configuration, every choice described in this chapter, was frozen and committed to version control **before any deep model was trained**. An adversarial review of the training code then produced a set of corrections, committed before any re-run and informed by no test result. Chapter 4 reports every configuration that informs a conclusion and states the total number explored, with the complete run record archived alongside the project, because with 58 subjects the optimism introduced by selective reporting is real and cannot be engineered away.

3.9 Confound-Control Protocol

Every model, classical or deep, is read against the same set of controls, fixed before training:

- **Null floors.** Two deliberately impoverished predictors are scored under the identical protocol: recording duration alone, and the window count alone. Any model not clearly above these floors has not demonstrated anything about gait.
- **Duration controls.** The one clinically real but unspecific difference, PD subjects taking longer, is controlled by construction through the fixed-length windows of Section 3.3, exposed by the `duration_only` floor reported beside every model, and tracked by correlating each model's subject scores with its window counts, so that elapsed time cannot masquerade as a spectral finding.
- **Uncertainty at the subject level.** Confidence intervals are obtained by bootstrap resampling of *subjects*, never of recordings or windows.
- **Attention verification.** Gradient-based attention maps (Grad-CAM) are computed for both image networks to verify which image regions drive their output, the direct check on shortcut learning that the reviewed literature lacked; they are reported alongside the discussion.

3.10 Evaluation Metrics

Every subject leaves the validation loop with a probability of being PD. Turning probabilities into decisions requires a threshold, and any single threshold is a contestable choice; the headline metric therefore avoids the choice entirely. The **area under the**

ROC curve (AUC) has a direct probabilistic reading:

$$\text{AUC} = P(s_{\text{PD}} > s_{\text{control}}), \quad (3.7)$$

the probability that a randomly chosen PD subject receives a higher score than a randomly chosen control subject; 0.5 corresponds to a coin flip and 1.0 to perfect ordering. Three properties make it the right headline here:

- It needs **no threshold**;
- It is **unaffected by the 57/43 class imbalance**, which lets a trivial classifier reach 57 % accuracy;
- It measures **ranking**, which is what a screening tool does when it flags who deserves closer examination.

Balanced accuracy, sensitivity and specificity are reported alongside, with the operating threshold set at the prevalence of the positive class rather than 0.5, since a model trained on an imbalanced dataset is not calibrated and a half-way cut systematically under-predicts the smaller group; the AUC itself is independent of this choice. Finally, a caution that recurs in Chapter 4: the literature this thesis is compared against overwhelmingly reports *accuracy* on random splits, which is not the same quantity measured under the same conditions, and a higher number obtained that way is not automatically the better result.

4. Results

4.1 Dataset Characterisation and Exploratory Analysis

This section reports what the dataset contains before any model is trained. It is presented as a result rather than as preparatory work, because its findings **constrained the design of everything that follows**: they changed which question could honestly be asked, how any answer must be validated, and which features can be trusted.

4.1.1 Cohort composition

Figure 4.1 shows the composition. The class imbalance is mild, **33 control against 25 PD**, or roughly 57 % to 43 %. It is small enough not to require resampling, but large enough that plain accuracy would mislead: a classifier that always predicted the majority class would score 57 % while being diagnostically useless. This motivates the use of **area under the ROC curve** as the headline metric, supported by balanced accuracy, sensitivity and specificity.

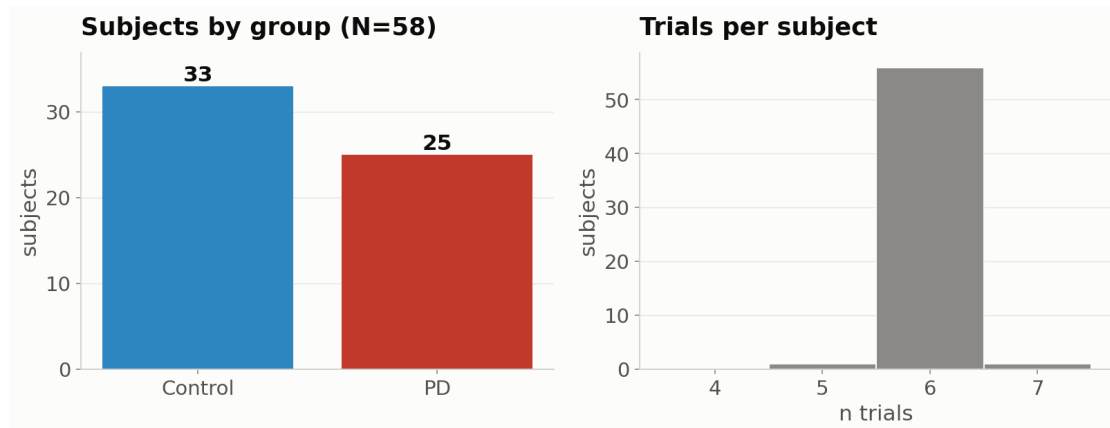


Figure 4.1: Subjects per group, and the number of recordings contributed by each subject.

4.1.2 Recording duration differs between groups, and the chair is why

PD subjects take longer to complete the test, and the difference is **concentrated almost entirely in the variant that includes the chair**, as Figure 4.2 and Table 4.1 show.

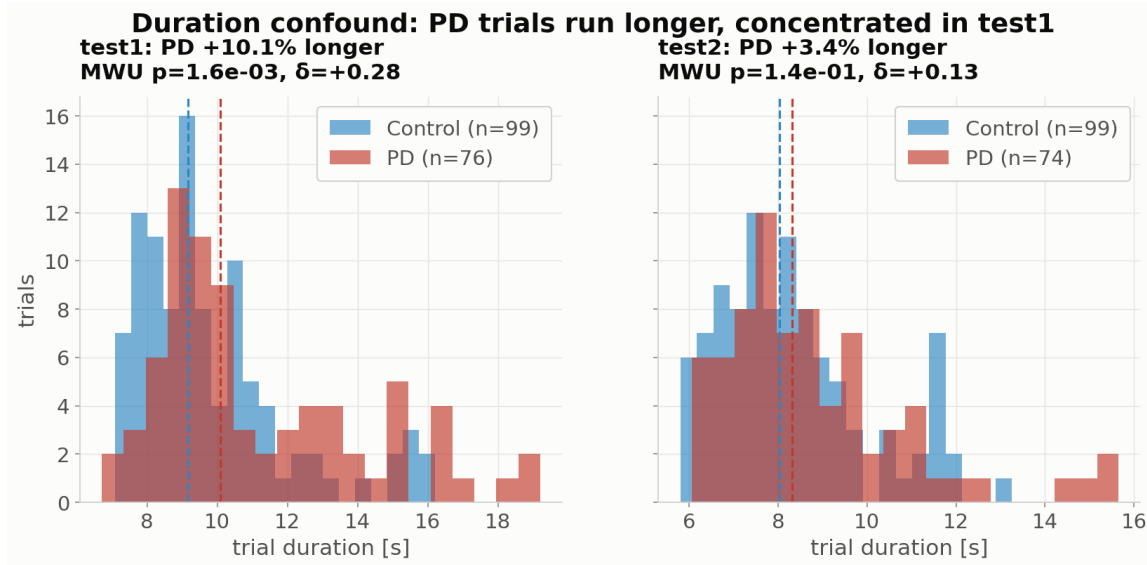


Figure 4.2: Distribution of recording durations for each group, split by test variant. Dashed lines mark each group’s median.

Table 4.1: Recording duration by group and test variant. δ is Cliff’s delta; p is the Mann-Whitney probability value.

Variant	Control median	PD median	Difference	p	δ
test1 (with chair)	9.17 s	10.09 s	+10.1 %	0.0016	0.278
test2 (without chair)	8.04 s	8.31 s	+3.4 %	0.141	0.131

In test1 the gap is unlikely to be coincidence and of small-to-moderate magnitude. In test2 it falls to a negligible effect size entirely consistent with ordinary between-subject variation. **Removing the chair removes most of the difference**, which is the behaviour predicted by the clinical picture: rising from a chair is among the movements most affected by bradykinesia [1], so the additional seconds accumulate in the sit-to-stand and stand-to-sit transitions rather than during steady walking. The same comparison was reproduced through an independently written analysis path and reached the same conclusion.

Two consequences follow, and both are carried through the rest of this thesis:

- **Elapsed time is a confound, not a feature.** Any method that responds to recording length will appear to detect the disease. A `duration_only` control model is therefore evaluated alongside every other feature set, so that its score acts as a floor below which no result can be considered evidence of gait.
- **The chair-free variant, test2, is the cleaner comparison.** With the chair removed the two groups take almost the same time, so a difference found there cannot be explained by one group simply spending longer in front of the radar.

4.1.3 Nearly every feature differs, which is a warning rather than a result

Comparing the two groups across all 25 features gives a result that appears excellent: **24 of the 25 show a difference unlikely to be coincidence** after correction for multiple comparisons. It is in fact a warning sign.

It is not plausible that a walk differs between the groups in 24 independent ways. It is very plausible that **one shared factor differs** and that it surfaces in nearly every measurement, and Section 4.1.2 has already identified that factor. Figure 4.3 confirms the diagnosis directly.

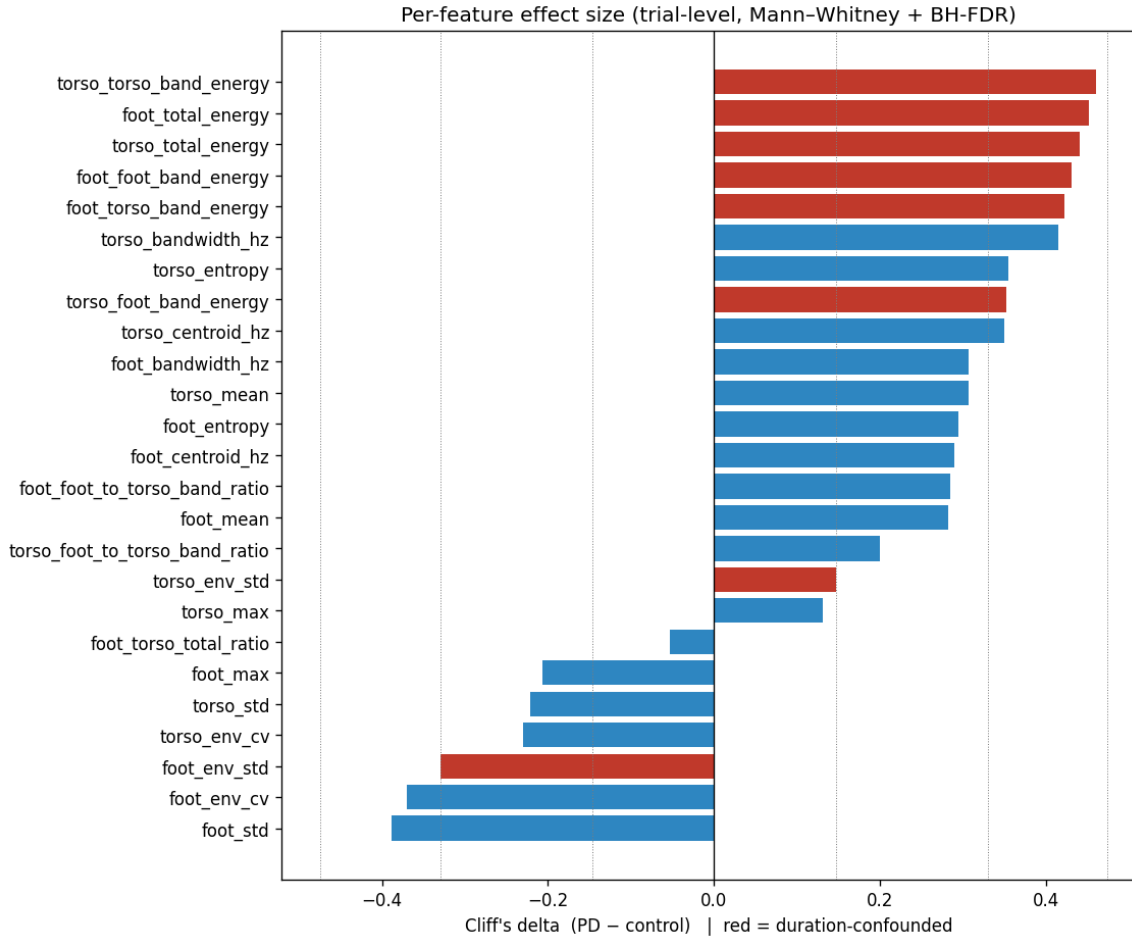


Figure 4.3: Cliff's delta for each of the 25 features, sorted. Red marks the features identified in advance as growing with recording length; blue marks the remainder. Dotted guides mark the negligible, small and medium thresholds.

Two observations stand out:

- **The five largest effects are all length-dependent.** Ranked by magnitude, the top five are all summed-energy features, and the first length-independent feature ranks sixth at $\delta = 0.41$. The strongest apparent differences therefore come from features that partly encode elapsed time rather than the shape of the motion.

- **No feature reaches a large effect at all.** The largest value in the dataset is $\delta = 0.461$, marginally below the conventional threshold for a large effect [23], and it is one of the length-dependent features.

The general lesson is that **a large number of significant results is weak evidence, not strong evidence**: it indicates that the measurements are not independent. Two or three differing features would have been more convincing, because that pattern would suggest something specific rather than something that touches every measurement.

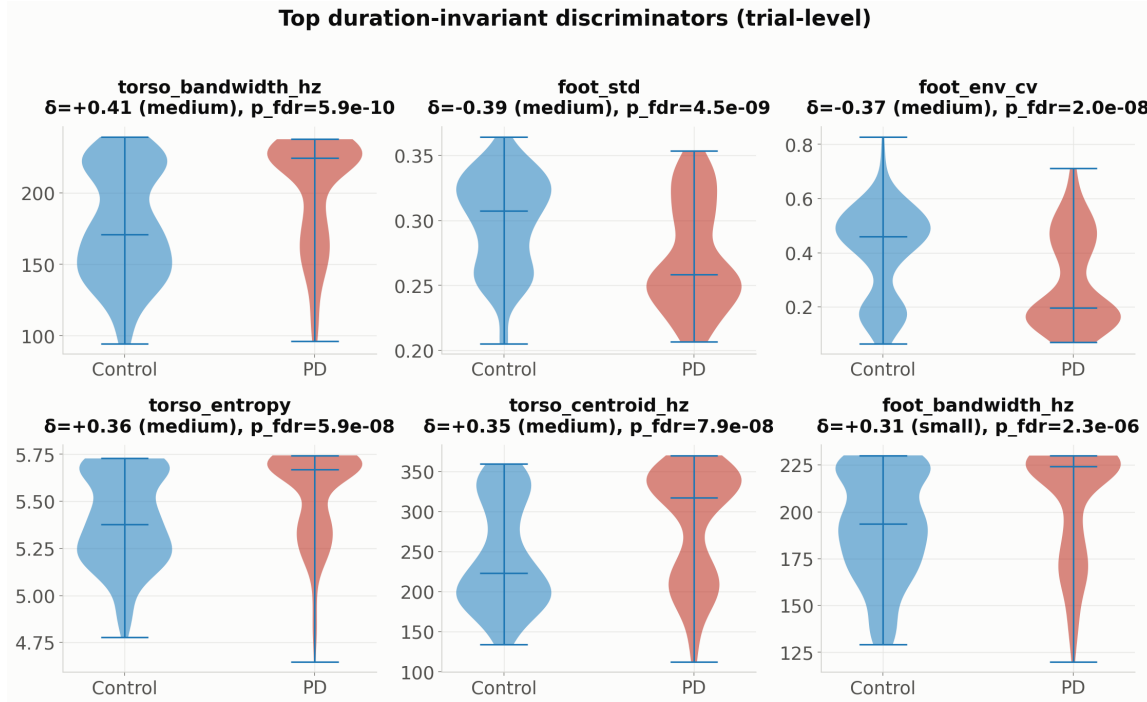


Figure 4.4: Distribution of each group's values for the strongest length-independent features. Wider regions indicate more recordings at that value.

Figure 4.4 shows what a probability value cannot. A small probability establishes that the groups differ **on average**; it says nothing about whether an **individual** could be assigned to the correct group, which is the only question a diagnostic tool has to answer. The distributions answer it directly: even for the best features, the two groups **overlap almost completely**. The value read from a PD subject is a value that many control subjects also produce.

This observation sets the expectation for the modelling results reported later in this chapter. Those results are modest, and **this overlap is the reason**. It was visible in the dataset before any model was built, which makes it a property of the problem rather than a shortcoming of a particular method.

4.1.4 A walk identifies the walker

Every subject walks the same 3 m corridor, so the following analysis concerns similarity rather than physical distance. Using the dissimilarity of Equation (3.6), each subject yields two numbers: the **within-subject** distance among their own recordings, and the **between-subject** distance to other subjects' recordings. If a walk carried no personal signature, the two would be equal.

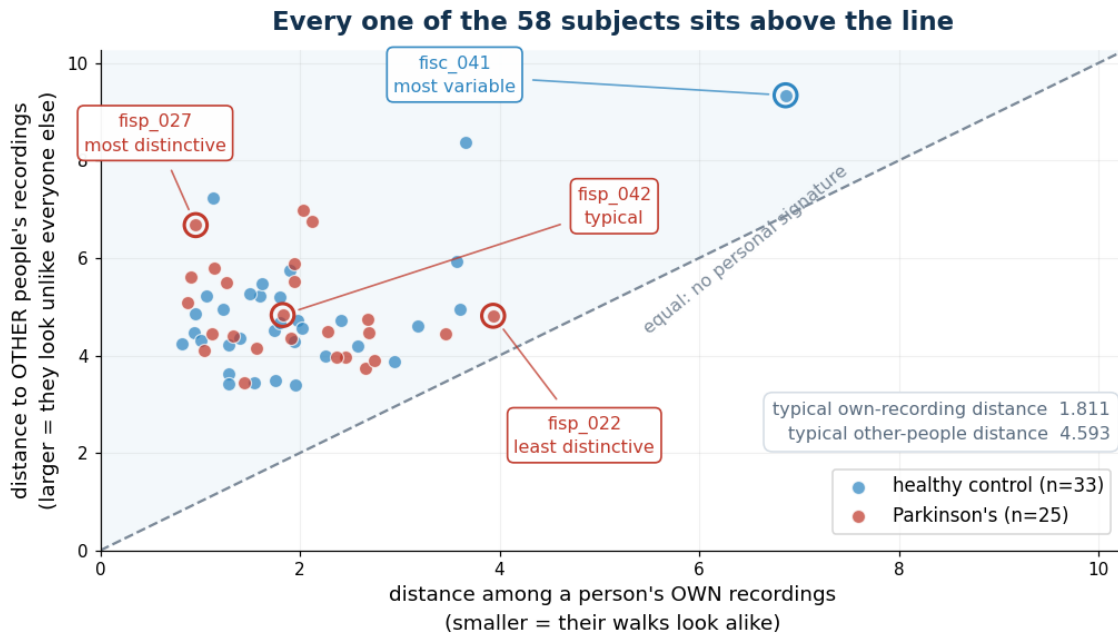


Figure 4.5: One point per subject. The horizontal position is the mean distance among that subject's own recordings; the vertical position is the mean distance to ten other subjects' recordings drawn at random. The dashed line marks equality. Four subjects are ringed and named to indicate the range observed.

The result, shown in Figure 4.5, is unambiguous. Every one of the **58 subjects lies above the line**, with no exceptions. A subject's own recordings sit a median distance of 1.81 apart, against 4.59 to other subjects' recordings, so a subject's own walks are approximately **2.5 times more alike** than anyone else's.

The interpretation requires care. The quantity being tested is not consistency: the subject with the widest spread among their own recordings, at 6.87, still lies above the line because they sit 9.34 from everyone else. What makes a subject identifiable is being **more similar to themselves than to others**, which is a comparison rather than a property of an individual.

This is a **hazard rather than a useful finding**. Were one recording of a PD subject used for training and a second recording of the same subject used for evaluation, a model would not need to learn anything about Parkinson's disease: it could recognise **the individual** and return the label already associated with them. The resulting score would be genuine and the conclusion drawn from it would be false. Two rules follow, and both are applied throughout:

- Every subject is held out **in full**, never one recording at a time, which is the **leave-one-subject-out** protocol of Section 3.8;
- Uncertainty is estimated by resampling **whole subjects** rather than individual recordings, because six recordings of one subject are not six independent observations and treating them as such would understate the uncertainty.

4.1.5 How the exploratory analysis constrained the design

Three elements of the original plan had to be revised once the dataset had been examined. Table 4.2 records them, because the revisions are themselves a contribution of this thesis.

Table 4.2: Design decisions revised by the exploratory analysis.

Planned	Found	Revised
Measure how accurately a model separates Parkinson's from healthy control.	PD subjects take longer to complete the test.	A model responding only to elapsed time would score well having learned nothing about gait. The objective became measuring how much separation survives once duration is removed .
Split the recordings at random into training and test sets.	Two recordings of the same subject are far more alike than recordings of two different subjects.	A model could score well by recognising the individual . Splitting is performed by subject , never by recording.
Decide by inspection which features grow purely with recording length, and exclude them.	Those judgements were wrong in both directions; several features assumed to be safe track duration closely.	Membership of the length-independent set is determined by testing each feature against measured duration .

The third revision has a cause worth stating, because it is not obvious. A longer recording is not the same walk performed more slowly. It contains proportionally **more standing, hesitating and turning** and proportionally less steady walking, so the mixture of activity inside it differs. A mean, which does not accumulate with length in the way a sum does, therefore still shifts when a recording is longer, which is exactly why membership of the length-independent set cannot be settled by inspection.

4.2 The Null Floors

Every model in this chapter is judged against the two deliberately impoverished predictors of Table 4.3, scored under the identical leave-one-subject-out protocol of Section 3.8. Any model not clearly above them has demonstrated nothing about gait.

Both floors carry **no gait content whatsoever**, yet both sit above 0.6. They quantify, before any model is discussed, how much apparent performance the recording-length confound of Section 4.1.2 supplies for free.

Table 4.3: The null floors.

Floor	AUC	How it is computed
Recording duration alone	0.610	One number per subject, how long their recordings last, fed to a logistic regression under the same leave-one-subject-out loop; the <code>duration_only</code> cell of Table 4.4.
Window count alone	0.620	No model at all: subjects are ranked directly by how many windows they yield (a restatement of duration, Section 3.3) and the AUC of that ranking is computed.

4.3 Classical Baselines

Table 4.4 reports the full grid promised in Section 3.6: three classifiers by five feature sets, each cell the subject-level AUC of a complete leave-one-subject-out run, drawn as a heat map in Figure 4.6.

Table 4.4: Classical baseline AUCs: three classifiers by five feature sets (feature sets as defined in Table 3.7).

Model	<code>all_features</code>	<code>duration_invariant</code>	<code>shape_clean</code>	<code>confounded_only</code>	<code>duration_only</code>
Logistic regression	0.633	0.627	0.621	0.675	0.610
SVM	0.623	0.601	0.577	0.634	0.287*
Random forest	0.605	0.601	0.583	0.618	0.450

*A one-feature kernel model can score below chance under leave-one-subject-out because its probability calibration is refitted in every fold, which makes the pooled scores incomparable across folds and scrambles the subject ranking; the anomaly is a calibration artifact of that cell, not a property of the data, as the logistic regression on the same single feature (0.610) shows.

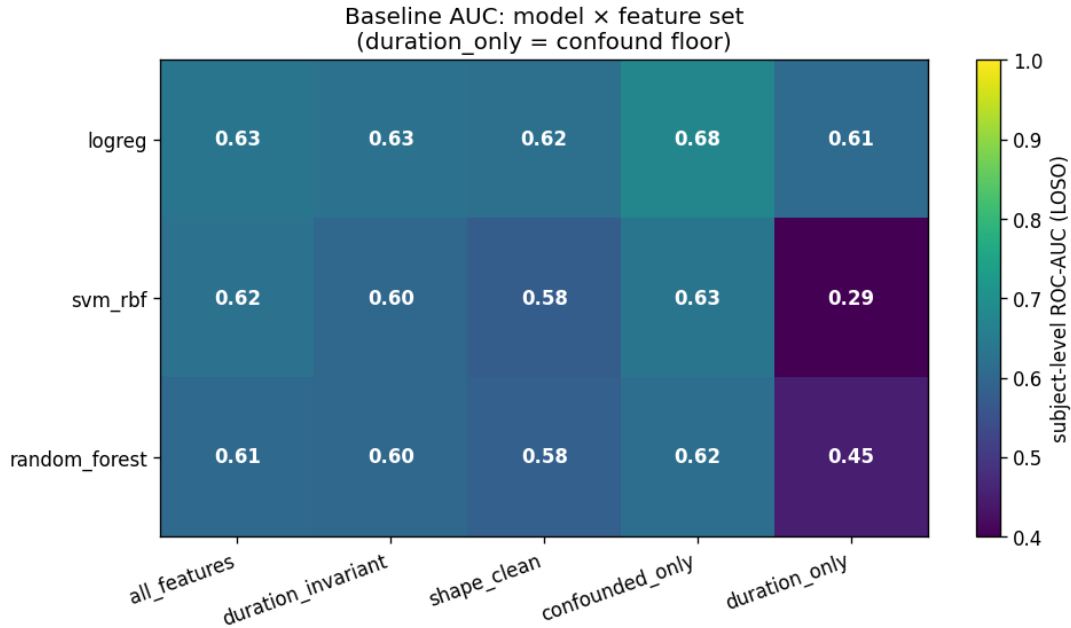


Figure 4.6: The grid of Table 4.4 as a heat map. Every row peaks in the `confounded_only` column: the warmth gradient runs toward the features that encode recording length, not away from them.

The logistic regression row is read **right to left**. Its `duration_only` cell, recording length alone, scores 0.610. Its `confounded_only` cell, the measurements that merely encode length, scores **0.675**, the best cell in the table. Across the four multi-feature sets the ordering follows the duration content exactly: 0.675, then 0.633, 0.627 and 0.621 as duration information is progressively removed. **The more duration information the features carry, the better the classifier does:** the grid is the controlled experiment Section 3.6 designed, and it came out in the direction that indicts the confound.

The bootstrap confidence intervals of Section 3.9 say which of these numbers is real. The interval on the confounded cell runs from **0.527 to 0.813** and excludes 0.5: the length signal is genuinely there. The interval on the clean cell runs from **0.473 to 0.770** and contains 0.5: a gait signal has **not been shown to exist**.

4.4 Deep Models

Table 4.5 reports the three models of Section 3.7 under the full nested protocol, and Figure 4.7 places them, together with two of the exploratory variants of Section 4.5 (foot channel only and turn excluded), against the floors.

Table 4.5: Deep models under leave-one-subject-out validation.

Model	Pooled AUC	95 % CI
SmallCNN	0.616	[0.468, 0.760]
ResNet-18 probe	0.654	[0.509, 0.798]
Envelope-LSTM	0.560	[0.404, 0.714]

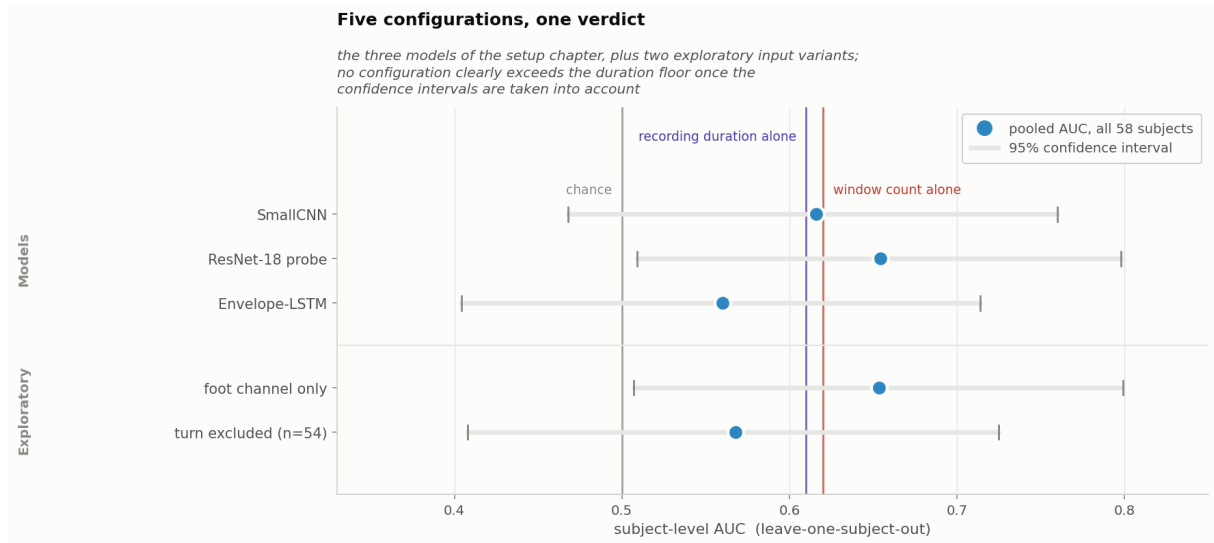


Figure 4.7: The three models and two of the exploratory input variants. Dots are pooled subject-level AUCs, bars their 95 % confidence intervals; the vertical lines are chance and the two floors of Table 4.3.

Three observations carry the section:

- **Every confidence interval contains the duration floor.** No configuration is shown to exceed a predictor that reads only how long the walk took. Two intervals (the probe and the foot-only variant) clear chance; none clears the floor.
- **More capacity and transfer learning do not change the picture.** The scratch-trained SmallCNN (23,682 parameters), the ImageNet probe (1,026 trainable), and the sequence model see the problem through entirely different representations and land in the same band.
- **The representation that cannot see image texture scores lowest.** The Envelope-LSTM, which receives velocity curves rather than images and is therefore blind to per-recording texture, reaches only 0.560, and in 28 of its 58 folds no epoch beyond the first improved on the first epoch's result. The most economical reading is that what the image models add over it is texture rather than gait, a reading the attention analysis of Chapter 5 tests directly.

4.5 Ablation Studies

Table 4.6 reports four input variants, each a re-run of the SmallCNN under the full pipeline with exactly one component changed (Section 3.3; the raw-magnitude variant instead bypasses the enhanced representation of Section 3.1). Each answers a design question raised in the methods chapter, and the first two also appear in Figure 4.7.

In total, eleven deep configurations were explored across the campaign and these variants. Those that inform a conclusion are reported in this chapter, and the complete run record is archived with the project, so no configuration was selected for its score.

Table 4.6: Exploratory input variants (full protocol, pooled AUC).

Variant	AUC	What it answers
Foot channel only	0.653	The source publication's prediction that the foot channel carries more usable signal. A first pass reached 0.674; the pre-registered re-test settled at 0.653, indistinguishable from the two-channel models. No channel advantage survived.
Turn excluded ($n = 54$)	0.568	Whether the signal lives in steady walking. Restricting to single-direction windows lowers the score toward chance: the models were reading the turn-and-stand content, where the group time difference of Section 4.1.2 lives.
Per-window standardisation	0.647	Whether absolute brightness hides a signal or a nuisance. Deleting it removes most of the length residue and uncovers nothing beneath: nuisance.
Raw STFT	0.661	What the contrast enhancement contributes. Bypassing it restores absolute energy and the scores track recording length twice as strongly: the rise is the nuisance channel, not gait.

4.6 Decision Performance at a Fixed Threshold

AUC measures ranking; a screening deployment must decide. Following Section 3.10, each model’s per-subject probabilities are cut at the prevalence of the positive class ($25/58 = 0.431$), and Table 4.7 reports the resulting decisions for the classical baseline and the three deep models, and Figure 4.8 draws the four confusion matrices.

Table 4.7: Decision metrics at the prevalence threshold (positive class: PD).

Model	Recall	Specificity	Precision	F1	Bal. acc.
Logistic regression (<code>shape_clean</code>)	0.68	0.55	0.53	0.60	0.61
SmallCNN	0.60	0.67	0.58	0.59	0.63
ResNet-18 probe	0.68	0.48	0.50	0.58	0.58
Envelope-LSTM	0.60	0.64	0.56	0.58	0.62

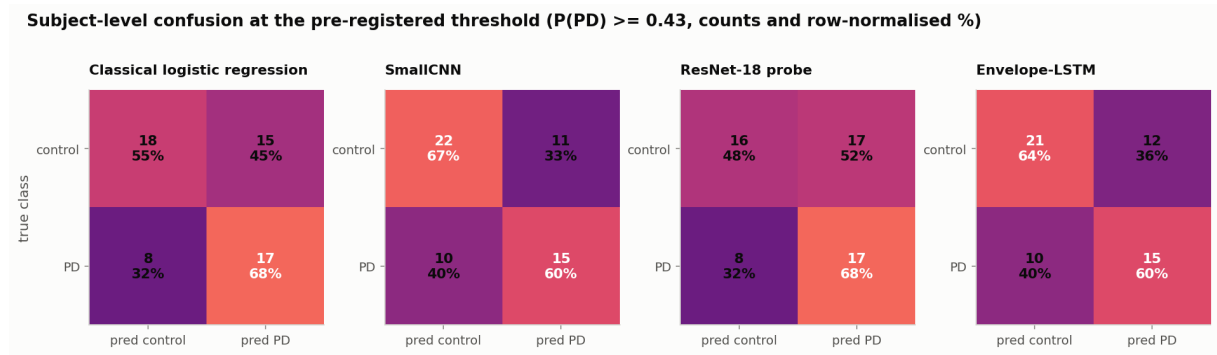


Figure 4.8: Subject-level confusion matrices at the prevalence threshold, one per model: counts and row-normalised percentages.

At this threshold the best deep model catches **17 of the 25 PD subjects**, misses 8, and falsely flags **17 of the 33 controls**; balanced accuracy spans 0.58 to 0.63 across the four models. The per-subject view of Figure 4.9 shows why no other threshold would do better: the two classes are interleaved along the entire probability ramp, so sliding the line only trades one kind of error for the other. The weak decisions are a property of the scores, not of the cut.

Figure 4.10 attaches a concrete case to each cell of the confusion matrix, and the pattern is legible in the backgrounds rather than the strides: for both models, the subjects **called PD** (correctly or not) sit on bright, noisy backgrounds, and the subjects **called control** (correctly or not) on dark, clean ones. The most extreme case is the probe’s false alarm, a control flagged at 0.81 whose window is almost pure noise texture, beside a genuine PD subject waved through at 0.24 on a clean background. The errors are not random; they are the systematic price of a score that follows recording character rather than gait, an observation taken up in the discussion.

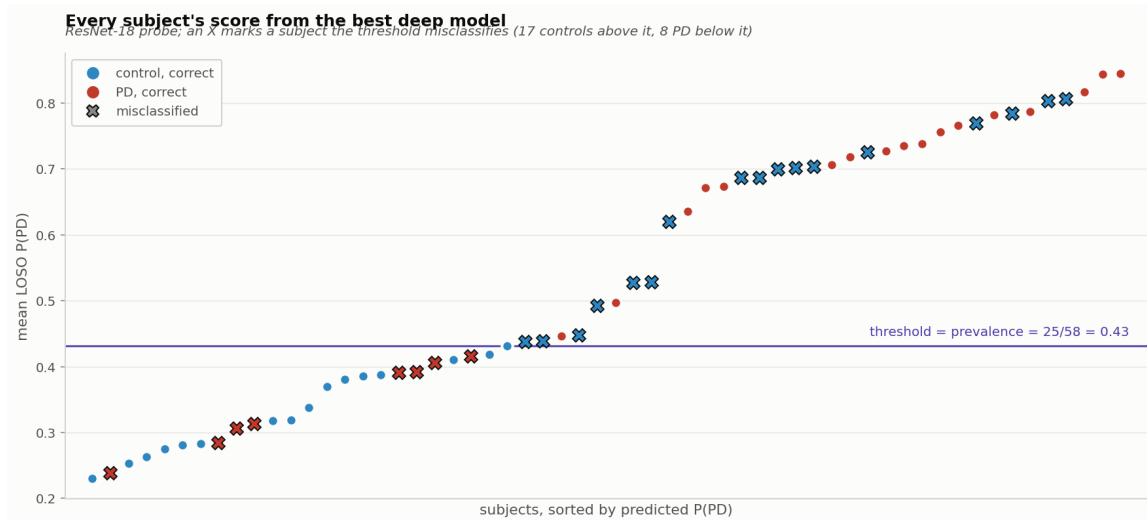


Figure 4.9: Per-subject probabilities of the ResNet-18 probe, sorted. The violet line is the decision threshold at 0.431, the cohort's own PD share; an X marks each misclassified subject.

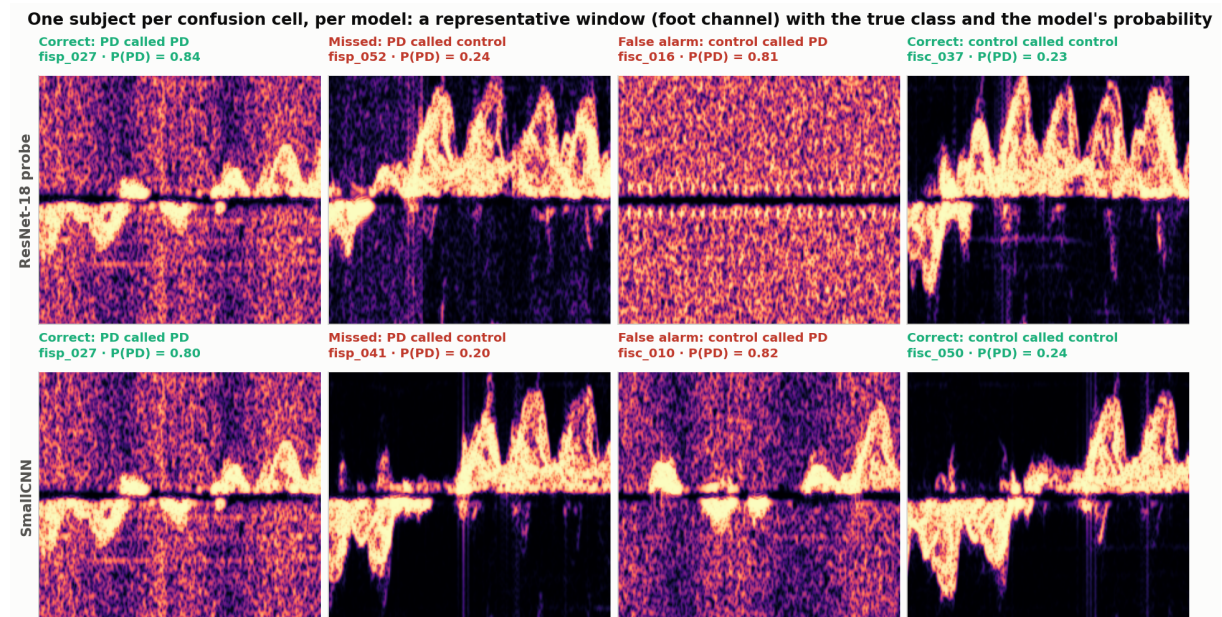


Figure 4.10: One subject per confusion cell for the two image models: the most confident correct PD, the most confidently missed PD, the most confident false alarm, and the most confident correct control, each shown through a representative window (foot channel) with the model's probability.

5. Discussion, Conclusions and Future Work

5.1 Discussion

The results chapter measured what this dataset supports under subject-independent validation: subject-level scores that sit in the same band as the score recording duration provides on its own (Section 4.2). Two questions follow from that measurement, and this chapter answers both with experiments rather than argument. The first is why published radar gait studies report so much more. Section 5.1.1 reproduces the literature’s numbers on this very dataset by changing nothing but the validation design, first with the campaign’s own network and then with the field’s favourite architecture, and shows that those numbers collapse the moment a test subject is genuinely new. The second is what the scores obtained here are actually made of. Section 5.1.3 reads the models’ own attention maps, Section 5.1.4 deletes what they attend to and confirms that no hidden gait signature was underneath, and Section 5.1.5 states the findings the data does establish. Section 5.2 closes with the limitations; the concrete path to stronger results follows in the future-work section.

The framing this chapter defends can be stated in advance. The gap between this thesis’s numbers and the literature’s is not a performance gap; it is a **measurement gap**, because the two are answering different questions. The literature’s question, whether a model can label recordings of people it has already seen, is answered here too, and with literature-grade numbers. The clinical question, whether a model can flag a person it has never seen, is the one this thesis insists on, and its honest answer on this dataset is: not yet, and not demonstrably from gait shape.

5.1.1 The random-split demonstration: the gap is the protocol

The claim that the literature’s numbers are inflated by validation design would normally rest on reading papers. Here it was tested directly: the same network and data as the main campaign, trained the field’s way, with a random hold-out split, early stopping on window-level scores and no window weighting, so that the whole published recipe is reproduced rather than argued about. Table 5.1 reports the outcome.

Table 5.1: One architecture under four validation designs. All rows use the ResNet-18 probe of Section 3.7; the first three are trained identically and differ only in what the split keeps apart.

Validation design	AUC	What it measures
Window-level random split	0.774	Nearly identical overlapping windows of the same walk sit on both sides of the split (3 repeats).
Recording-level random split (the literature’s hold-out)	0.760	Every test recording comes from a person the model trained on (5 repeats).
Same trained model, subjects excluded from training	0.579	The person is unknown; only the disease could help (3 repeats).
Leave-one-subject-out (the thesis protocol)	0.654	Every subject scored by a model that never saw them (subject-level AUC, Table 4.5).

The rows deserve unpacking, because the mechanics decide what each number means. In the **window split**, the three-second windows themselves are shuffled into training, validation and test piles. In the **recording split**, whole walks are shuffled instead, so each test walk is new but the walker’s other walks are in training; this is the design most published studies use. In the **third row**, eight subjects (five control, three PD) are set aside before anything is trained; the model trains recording-split style on the remaining fifty subjects, and the finished model then classifies every window of the eight strangers’ recordings, with a fresh eight in each repeat. In the **last row**, one subject is set aside, the model trains on the other 57, and scores every window of all of that person’s recordings; the average of those window probabilities becomes that one person’s score, and repeating this 58 times gives the subject-level AUC of the main campaign. One unit caveat applies: the first three rows are window-level AUC averaged over the repeats, the level at which the literature scores, while the last row is subject-level, one score per person. The trend is the point: the more the split leaks, the higher the number, and stripping the leakage returns the same model, on the same data, to the modest band reported in Chapter 4.

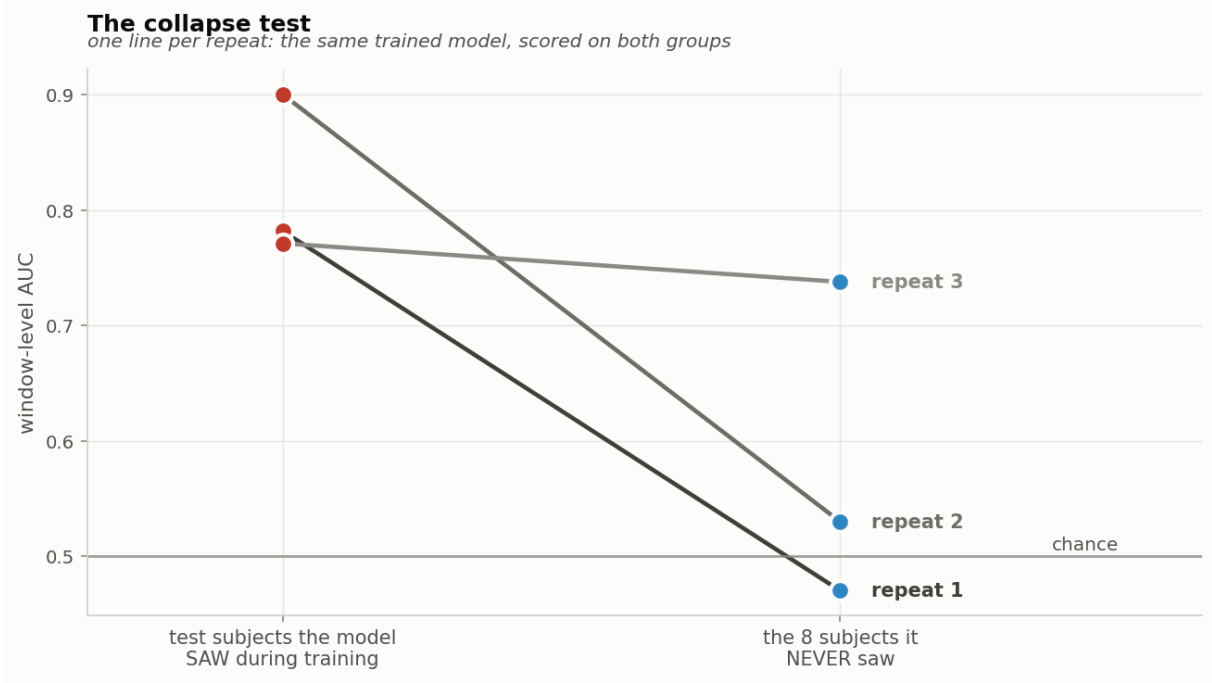


Figure 5.1: The collapse test, the third row of Table 5.1: three instances of the ResNet-18 probe, one line each. The left end of each line is that model’s window AUC on test recordings of subjects it saw in training; the right end is the same model on the eight subjects excluded from training.

Each line in Figure 5.1 is one trained model, not one subject. On the left, every model scores high on test recordings of people whose other walks it trained on; on the right, the very same models are asked about the eight people they have never seen, and every line falls, with two of the three landing at chance. A model that had learned the disease would hold its score on strangers, because Parkinson’s looks like Parkinson’s in people it never met. A model that learned **people** scores high on the familiar and collapses on the unknown, and that is what happens. The inflated numbers are real numbers; they are simply answers to the recognition question, not the screening question. Once it is established that the validation design alone manufactures a difference on the order of 0.2 AUC on this very dataset (0.18 to 0.20 between the averaged rows of Table 5.1, and up to 0.4 in the most extreme repeat of Figure 5.1), comparisons with published accuracies stop being embarrassing and start being diagnostic.

5.1.2 The same demonstration with the field’s own network

The demonstration above uses the thesis’s own network, and a sceptic could object that the collapse says something about that particular model rather than about the protocol. The experiment was therefore repeated with **AlexNet**, the architecture behind the field’s headline micro-Doppler result [7], given the same input adaptation as the ResNet-18 probe of Section 3.7: ImageNet-pretrained, with the first convolution averaged to the two radar channels. Unlike the probe, which retrain a single 1,026-parameter layer, AlexNet’s full classifier block is retrained (54.5 million trainable parameters) at the source literature’s learning rate. The window cache and augmentation are those of the main campaign; the random-split runs additionally adopt the literature’s training conventions, early stopping

on window-level scores and no window weighting. Table 5.2 reports the outcome under both designs.

Table 5.2: AlexNet under the literature’s protocol and under the thesis protocol. A seed is one repeat of the split: the same recordings shuffled with a different random starting point.

Validation design	Train / val / test	Subject AUC	95 % CI	Seen in training
Random split, seed 1	70/15/15 % recordings	0.875	–	100 %
Random split, seed 2	70/15/15 % recordings	0.891	–	100 %
Random split, seed 3	70/15/15 % recordings	0.735	–	100 %
Leave-one-subject-out	49/8/1 subjects, $\times 58$	0.695	[0.554, 0.833]	0 %

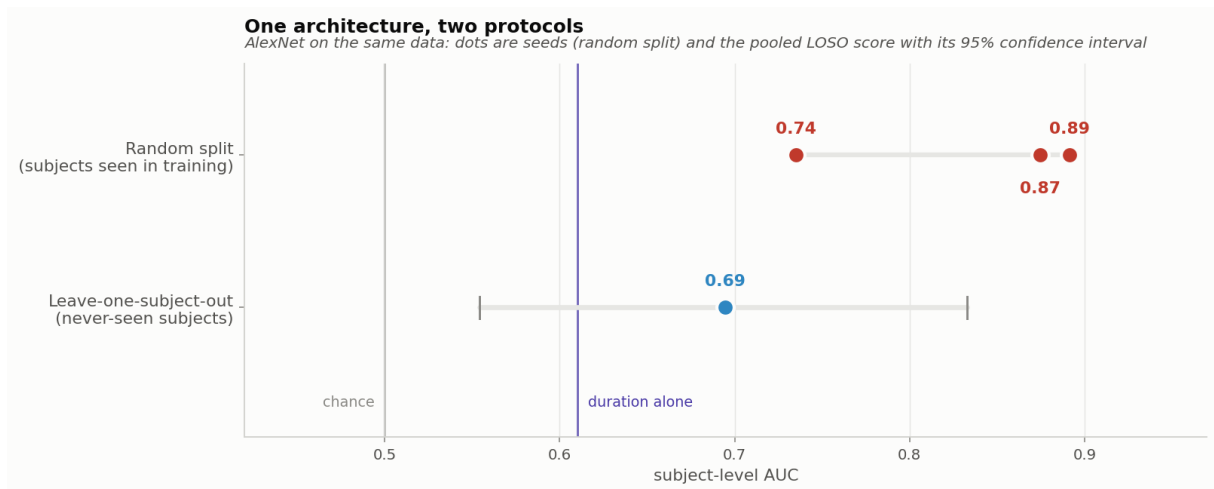


Figure 5.2: AlexNet’s subject-level AUC under the literature’s random split (red dots, one per seed, each labelled with its score) and under leave-one-subject-out (blue dot with its 95 % confidence interval), against the chance and duration-alone lines.

Figure 5.2 admits three readings. First, the field’s own architecture reaches 0.87 and 0.89 the moment every test subject is someone it trained on, and drops the moment they are not: **the split, not the architecture, decides** whether the number is literature-grade or honest. Second, the inflated number is itself unstable: reshuffling the same recordings moves it from 0.89 down to 0.74, a swing of 0.16 between seeds. Third, the honest interval reaches down past the duration floor of Table 4.3, so even the best protocol-compliant score in this project cannot be distinguished from what elapsed time alone provides.

An AUC summarises how well the scores *rank* people; a screening decision is a yes or no. Figure 5.3 therefore shows the same network’s actual decisions twice, first on subjects it saw in training and then on subjects it has never seen. The two panels count different things: on the left, each cell counts three-second windows, every test window classified on its own at the default 0.5 cut-off and pooled over the three seeds’ test sets; on the right, each cell counts people, with all windows of a person’s recordings scored, their average becoming that person’s single probability, and the person called PD when it exceeds the prevalence threshold of 0.431 (Section 4.6).

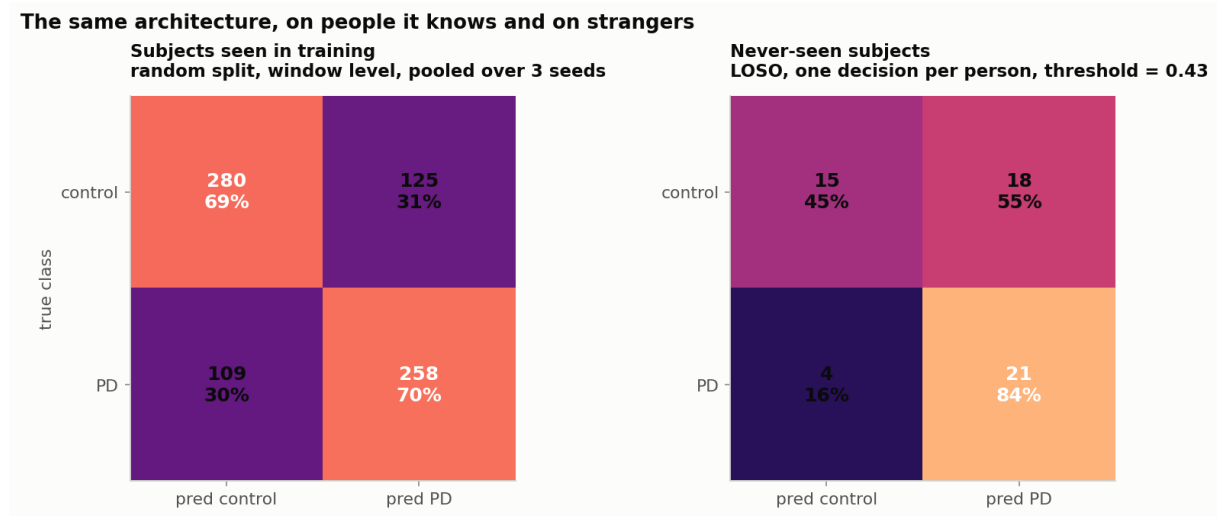


Figure 5.3: AlexNet’s decisions on subjects seen in training (left: random split, window level, pooled over the three seeds) and on never-seen subjects (right: leave-one-subject-out, one decision per person, threshold at the PD prevalence).

On seen subjects the model is balanced and looks useful, calling about 70 % of each class correctly. On never-seen subjects the balance breaks: it catches 21 of the 25 PD patients, but only by flagging 18 of the 33 healthy controls along with them, a specificity of 45 %, worse than a coin flip. The left panel is what the literature’s validation rewards, and it is real; it is also the answer to the recognition question, because every one of those windows belongs to a person whose other walks were in training. The right panel is the clinical question: faced with a stranger, the network can only lean toward calling recordings PD, so of the 39 people it would send onward for review, 18 are healthy. The same architecture is clinic-ready on people it knows and close to a biased guess on the population a screening tool actually exists for.

Two details make the honest AlexNet score itself instructive. Its subject scores carry **more duration residue than any of the campaign’s headline models** (correlation +0.25 with window count, against +0.08 for the ResNet-18 probe): with 54.5 million parameters free, it drinks more deeply from the same confound; it does not find a new signal. And its best validation epoch averages 0.9, meaning the score is essentially in place after a single pass over the data, the learning profile of an easy shortcut rather than of a subtle clinical pattern.

5.1.3 What the models actually read

The results chapter reported the scores; the attention analysis explains them. Grad-CAM maps [25] were computed for every window of each of the 58 held-out subjects, for both image models: the compact CNN trained from scratch (SmallCNN) and the pretrained deep network (the ResNet-18 probe). One gallery per model follows, so the finding can be checked on each.

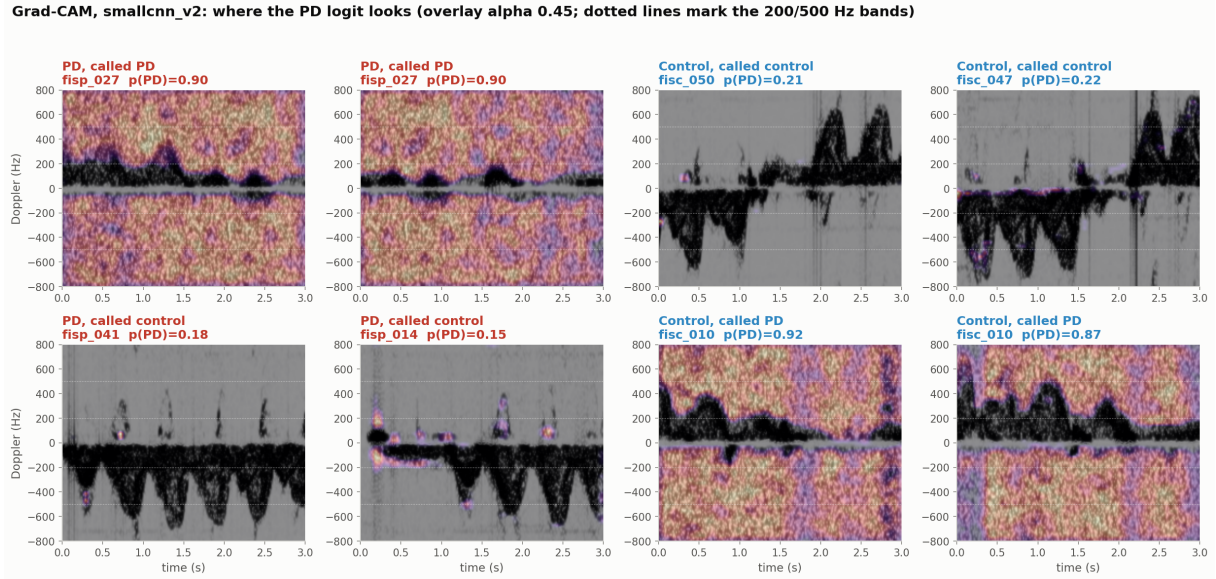


Figure 5.4: Where the SmallCNN's PD output looks, for confident correct decisions (top) and confident mistakes (bottom): spectrogram in grey, attention in colour.

In every window of Figure 5.4 called PD, right or wrong, the attention floods the empty background; the gait envelope itself is the one region left unhighlighted. In every window called control the map stays dark. The decision variable is a property of the recording, its background and enhancement texture, not of the walk. The mistakes obey the same rule, which is why they are systematic rather than random: noisy-backed controls get flagged, and clean-backed patients get missed, exactly the pattern already visible in the case gallery of Figure 4.10.

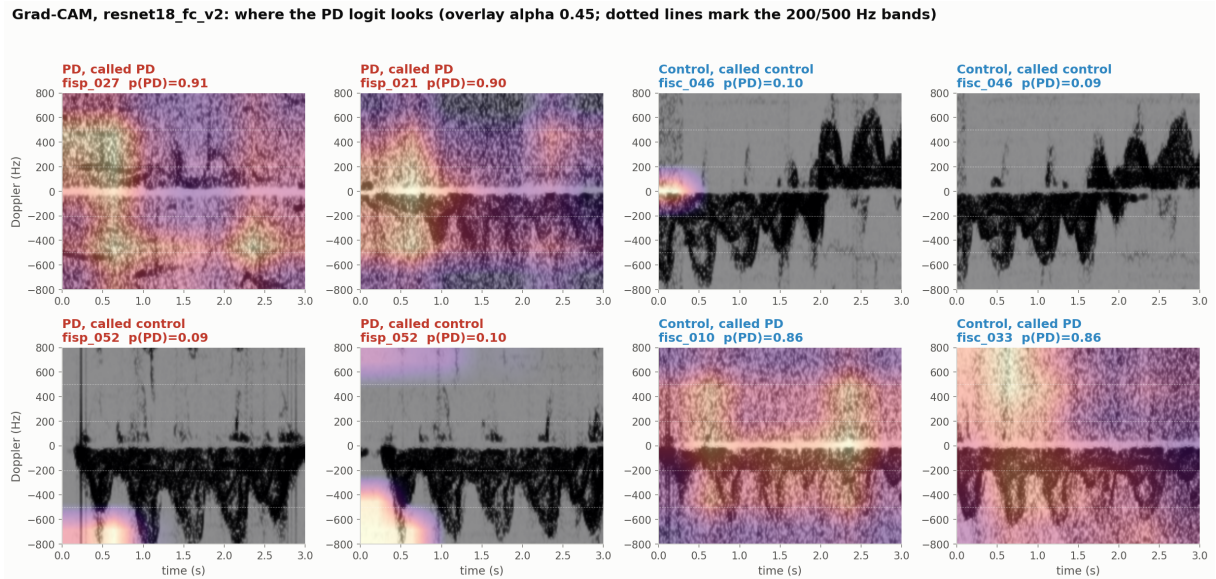


Figure 5.5: The same gallery for the ResNet-18 probe: confident correct decisions on top, confident mistakes below.

The deep network in Figure 5.5 obeys the same rule, expressed more diffusely: windows it calls PD are flooded with attention across the whole image, silent rows included, while

windows it calls control are left almost unattended. Averaged over every window of all 58 held-out subjects, its attention density on called-PD windows is 0.37 against 0.13 on called-control windows; for the SmallCNN the same pair is 0.33 against 0.04. Two different architectures, one trained from scratch and one built on frozen ImageNet features, converge on the same behaviour: respond to the overall character of the recording, not to any localised gait structure.

Three quantitative facts, computed over every window of all 58 held-out subjects, complete the picture:

- For the SmallCNN, attention density is highest in the silent rows above 500 Hz (0.19 per pixel against 0.12 in the torso band), a region containing no gait energy, and its attention sits visibly higher in frequency on PD windows than on control windows (centre of mass 380 against 333 Hz).
- For the SmallCNN, turn-straddling windows draw **four to six times** the attention of steady-walking windows, matching the ablation of Table 4.6 in which excluding the turn collapsed its score.
- The probe differs in one telling way: because its backbone is frozen, its maps average out **identical between the true classes** (background attention share 0.337 on PD windows against 0.338 on control windows). A fixed ImageNet description of these images contains nothing class-specific to point at, so its thin trained head can only read overall texture.

Taken together, the mechanism of every score in this thesis is now explicit: **elapsed time**, carried by turn-and-stand content and window counts, plus **per-recording character**, carried by background and enhancement texture. Both are properties a screening tool must not rely on; neither is the spectral shape of a Parkinsonian walk.

5.1.4 The intervention test: removing what they read

An interpretability finding invites an intervention: if the models read the background, delete the background and see what remains. The follow-up experiment, pre-registered before any window was rebuilt, subtracted each recording’s background profile and cropped the silent rows beyond 500 Hz, then re-ran both image models under the unchanged protocol. Table 5.3 reports the outcome.

Table 5.3: Both image models with the background suppressed (pooled AUC, full leave-one-subject-out protocol).

Model	Standard input	Background-suppressed	95 % CI (suppressed)
SmallCNN	0.616	0.624	[0.471, 0.768]
ResNet-18 probe	0.654	0.686	[0.539, 0.824]

Two conclusions follow, both informative:

- **No hidden gait signal surfaced.** If the texture had been masking a real gait signature, removing it should have lifted the scores clearly above the duration floor. Every confidence interval still contains 0.610.

- **The nuisance is distributed, not localised.** The scores did not collapse either: enough per-recording character survives inside the gait band, plausibly in the per-recording contrast enhancement, for the models to keep finding it. The recording-level nuisance is a property of how each file was processed, not one removable region of the image.

5.1.5 What was found, stated positively

Not everything in this thesis is a negative result. The findings of Table 5.4 are established by the data and hold under the strict protocol; each row gives the finding, the evidence in brief, and what it tells us.

Against the literature, these results are not an outlier; they are what the few honest data points predict. Of the sixteen works reviewed in Chapter 2, only two report a genuinely subject-independent protocol, and one of them saw its scores fall the moment it moved from a pooled split to leave-one-subject-out evaluation [15]. The study closest in modality to this work preferred its modest envelope model over a network trained on the full image whose higher score its own authors distrusted, because that network had keyed on background differences between recording sites [7]. This thesis extends that honest line to Parkinson’s screening and explains, mechanistically, where the distrusted scores come from.

Table 5.4: The positive findings.

Finding	Evidence	What it tells us
PD subjects take measurably longer , and the extra time concentrates in standing up and turning.	The duration gap between the groups is clear in the recording lengths, is largest in the walk variant that includes the chair phase, and shrinks when the turn is removed.	Radar captures the slowness of movement that clinicians already test for; the timing of a walk is a real, usable signal.
Radar timing is a credible screening quantity.	Recording duration alone separates the groups above chance under honest, subject-independent validation.	An automated timed-up-and-go measurement is this dataset’s trustworthy first-line signal, and it needs no deep network at all.
A walk identifies the walker.	Recordings of the same person are far more alike than recordings of different people, for every one of the 58 subjects (Section 4.1.4).	Subject-independent validation is mandatory, and radar gait could serve as a biometric even where diagnosis fails.
The field’s validation gap is real and measurable.	Relaxing the protocol on this one dataset reproduces literature-grade numbers that collapse on unseen subjects.	Published accuracies reflect validation design as much as clinical signal, so honest evaluation has to be demanded before numbers are compared.

5.2 Limitations

Four limitations bound what this thesis can claim. Table 5.5 states each one together with its consequence and the item that would resolve it.

Table 5.5: Limitations, and what would resolve each.

Limitation	Consequence, and what would resolve it
No age metadata.	The dominant untestable confound: PD subjects are typically older, radar separates young from elderly walkers at 94.9 % [7], and even the surviving duration signal could partly be ageing. Subject ages would allow an age-matched analysis.
Combined foot channels hide left/right asymmetry.	Asymmetry is an established PD marker, and it is structurally invisible after the two foot radars are merged into one channel. The original per-sensor recordings would make it measurable.
58 subjects.	With 58 people, every score carries a wide uncertainty range, so a small real effect can neither be proven nor ruled out at this size. Only a much larger group, in the hundreds, can settle small differences.
No clinical information beyond the label.	Each subject is only marked PD or control. Without disease stage, symptom severity or medication state, it is impossible to test whether the signal grows with severity, or how treatment changes it, both of which matter for a screening tool.

5.3 Conclusions

This thesis set out to determine whether Parkinson’s disease can be detected from the micro-Doppler gait signatures of a validated clinical radar network, and to do so under an evaluation a clinician could trust: every subject scored by a model that had never seen them. Five conclusions follow from the work; each notes, in parentheses, the objectives of Section 1.5 it closes.

A complete, pre-registered pipeline was built and frozen before the results were known (O2, O3). Preprocessing, windowing, model families, hyperparameters, validation and statistics were fixed in advance (Chapter 3), so every number reported in Chapter 4 is the first and only answer the protocol gave, not the best of many attempts.

Under subject-independent validation, no classifier was distinguishable from recording duration alone (O1, O5). Classical models, a compact CNN, the ImageNet probe, a sequence model on velocity envelopes, and the field’s own AlexNet all land in a band whose confidence intervals contain the duration floor of 0.610 (Sections 4.3, 4.4 and 5.1.2). The attention analysis and the background-suppression experiment attribute the remaining score to per-recording character rather than to gait shape (Sections 5.1.3 and 5.1.4).

The gap to the literature was reproduced, not just alleged (O5). With the validation design relaxed to the field’s random hold-out, the same data and networks produce subject-level AUCs up to 0.89 that collapse on people excluded from training (Sections 5.1.1 and 5.1.2). The published numbers answer a recognition question; the clinical question demands subject-independent evaluation, and this thesis provides, to the best of our knowledge, the first such evaluation for radar-based Parkinson’s classification.

One clinically sensible signal survives (O1). Patients take measurably longer, with the excess concentrated in standing up and turning; radar measures this timing reliably, which amounts to an automated timed-up-and-go measurement (Section 5.1.5). It is a modest signal, but it is real, interpretable, and it is exactly what the source system was validated to measure.

The negative result is the contribution (O4). This work demonstrates, on validated clinical hardware, precisely how radar gait classifiers come to look better than they are, provides the protocol that prevents it, and states which signals in this dataset are exhausted and which were never testable. Whatever signal richer data holds, the standard defended here will measure it honestly, on people the model has never seen.

5.4 Future Work

The limitations of Section 5.2 are not dead ends. Ordered from immediately actionable to long-term, Table 5.6 lays out how stronger and trustworthy results can be built on this line of work.

Table 5.6: Future work: the path to stronger results.

Step	What it would deliver
1. Model the timing signature directly.	Extract turn duration, sit-to-stand time and cadence from the micro-Doppler signature as explicit quantities, instead of hoping a network discovers them in pixels. Timing is the one signal this work showed to be real: patients take longer, and the extra time concentrates in standing up and turning. A model built directly on these timing quantities would be an automated version of the timed-up-and-go test that clinicians already use, which is also what the source radar system was validated to measure [6].
2. Obtain subject ages.	An age-matched analysis would settle whether the surviving signal is disease or ageing, and could also unmask real signal that age variation currently hides.
3. Record a larger, session-balanced cohort.	With 58 subjects every score carries a wide uncertainty range, so even a genuinely better model could not prove itself here. Demonstrating a spectral gait signature clearly above the duration floor needs a cohort in the low hundreds, recorded in sessions balanced across the two groups.
4. Keep the evaluation standard.	Whatever the model, evaluate it on people it has never seen and report it next to the duration floor and a confidence interval. The protocol built for this thesis is pre-registered, ablation-tested and reusable as the design document for every step above.

The review of this work also pointed to methodological improvements that target the specific failure found here, a network that looks at the wrong part of the image. Table 5.7 lists them, each with the work that introduced the idea.

Table 5.7: Future work: methodological improvements suggested at review.

Technique	What it would change
Make the network look at the gait.	Attention-guided masking trains the attention map itself to stay inside the region of interest [26]; explanation regularisation penalises attention on regions known to be irrelevant, so that the network is right for the right reasons [27, 28]; counterfactual training adds examples in which only the irrelevant part of the image changes, so that it cannot carry the label [29]. All three act directly on the failure the attention maps revealed.
Remove the background first.	An adaptive threshold set from the local energy of each signature strips the background before training and leaves only the moving-body content [30]. Combined with the masking above, it takes the shortcut away instead of asking the network to ignore it.
Learn a compact embedding first.	An encoder with a narrow bottleneck compresses each signature into a few informative numbers before any classifier sees it [31]. Plotting that embedding with t-SNE or UMAP [32, 33] shows at a glance whether it separates by diagnosis, by person or by recording length.
Borrow encoders from speech.	A micro-Doppler signature is closer to a speech spectrogram than to a photograph, and encoders pretrained on speech transfer to it better than image networks do [34, 35]; the idea goes back to speech-recognition front-ends applied to micro-Doppler [36]. This replaces the ImageNet probe with a source domain that matches.
Estimate the Doppler content parametrically.	An autoregressive spectral estimate replaces the short-time Fourier transform with a model fitted to each short segment, giving smoother signatures with less noise from the same data [37].
Tune the window length.	The 3.0 s window was fixed in advance from the stride duration. Treating it as a hyperparameter, chosen with the AUC as figure of merit inside the inner validation loop, would test whether a longer or shorter view helps.

5.5 Closing Remarks

This work asked a simple clinical question of a rich dataset and refused the shortcuts that would have made the answer look better than it is. What it hands to the next study is a modest and honest number, a clear explanation of why that number is not higher, and an evaluation protocol that any future attempt can be held to. The day richer data arrives, with ages, severity scores and the original per-sensor recordings, the standard defended here will be ready to measure it, on people the model has never seen.

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Annex A: Ethical, Economic, Social and Environmental Aspects

El apartado “Requisitos de las acreditaciones internacionales EUR-ACE y ABET” de la Normativa de TFT de la ETSIT-UPM establece que *“La memoria del TFT del GITST, GIB y MUIT, y en general la de aquellas titulaciones que hayan obtenido o para las que se desee solicitar una acreditación internacional EUR-ACE o ABET, debe mostrar conciencia de la responsabilidad de la aplicación práctica de la ingeniería, el impacto social y ambiental, el compromiso con la ética profesional, la responsabilidad y las normas de la aplicación práctica de la ingeniería, así como sobre las prácticas de gestión de proyectos, gestión, y control de riesgos, entendiendo sus limitaciones”*.

A1. Introducción

This project develops and evaluates a machine-learning system that screens for Parkinson’s disease from contactless radar recordings of gait. It touches three sensitive domains at once: **health data** from a vulnerable population, **clinical claims** produced by machine-learning models, and a sensing technology whose reach into private spaces is precisely its selling point. The relevant dimensions are therefore ethical (validity of clinical claims, privacy), social (access to earlier screening for an ageing population), economic (low-cost monitoring against expensive laboratory instrumentation) and, to a lesser degree, environmental (computational footprint).

A2. Descripción de impactos relevantes relacionados con el proyecto

Three impacts dominate, each with its stakeholders:

- **The validity of clinical machine-learning claims.** Stakeholders: patients, clinicians, the research field. An overstated screening accuracy harms twice: false alarms send healthy people to neurology consultations, and false trust delays real diagnoses. This thesis found that the field’s evaluation habits inflate performance substantially, and made honest evaluation its central contribution.
- **Privacy of gait data.** Stakeholders: monitored persons and their households. Radar records no image, which is its privacy advantage; yet this work also measured that a person’s walk identifies them (all 58 of 58 subjects distinguishable), so gait recordings are biometric data under the GDPR and must be treated as such:

anonymised identifiers, restricted access, purpose limitation. The dataset used here is fully anonymised and was collected under the clinical validation study of the source system.

- **Access and cost of motor assessment.** Stakeholders: health systems, patients in areas without movement-disorder specialists. A radar Timed-Up-and-Go, the one measurement this thesis validates as trustworthy, is cheap, contactless and automatable, and could extend objective motor assessment beyond specialist clinics.

A3. Análisis detallado de alguno de los impactos

The impact analysed in depth is the first: **what happens when clinical machine-learning performance is overstated**, because this thesis measured the mechanism directly. On this very dataset, the evaluation design used by most published work produces subject-level scores that look clinic-ready; scoring the same models on people excluded from training returns them to a band indistinguishable from a predictor that knows only how long the recording lasted. A deployment decision taken on the first number would field a screening tool whose real behaviour on new patients is close to a biased guess: when the field's own network was scored on unseen people (Section 5.1.2), catching 21 of 25 patients cost falsely flagging 18 of 33 healthy controls. The ethical posture adopted is therefore procedural rather than declarative: pre-registration of the analysis plan, subject-independent validation, explicit null floors beside every reported score, confidence intervals at the subject level, and publication of the negative result. This is the engineering-responsibility standard the EUR-ACE criteria ask the graduate to demonstrate, applied to the claim itself rather than to the artefact.

A4. Conclusiones

The project's sustainability profile is favourable: it consumes only modest computation (a single personal computer; no data-centre training), reuses an existing clinical dataset rather than collecting a new one, and its product is methodological, a protocol that prevents overstated clinical claims. The sustainability criteria added genuine value: treating the validity of the claim as the primary ethical object is what turned an apparently negative result into the thesis's contribution, and the privacy analysis (gait as a biometric) is itself one of the work's findings.

Annex B: Economic Budget

El apartado “Requisitos de las acreditaciones internacionales EUR-ACE y ABET” de la Normativa de TFT de la ETSIT-UPM establece que “*La memoria del TFT del GITST, GIB y MUIT, y en general la de aquellas titulaciones que hayan obtenido o para las que se desee solicitar una acreditación internacional EUR-ACE o ABET, ... debe incluir un presupuesto económico*”. La tabla siguiente presenta el presupuesto del proyecto. La mano de obra corresponde a las 360 horas del TFM (12 ECTS a 30 horas por crédito) a la tarifa de referencia de la plantilla; el único equipo empleado es el ordenador personal del autor (MacBook Pro 2021, chip M1 Max, 32 GB de memoria), amortizado por la fracción de su vida útil usada en el proyecto; la red de radar y el conjunto de datos fueron cedidos por el grupo de investigación sin coste directo para el proyecto, y todo el software empleado es de código abierto:

Table B.1: Presupuesto económico.

COSTE DE MANO DE OBRA (coste directo)		Horas	Precio/hora	Total
		360	15 €	5.400,00 €
COSTE DE RECURSOS MATERIALES (coste directo)	Precio de compra	Uso en meses	Amortización (en años)	Total
Ordenador personal: MacBook Pro 2021 (M1 Max, 32 GB)	3.200,00 €	7	5	373,33 €
Red de radar y conjunto de datos (cedidos por el grupo)	–	–	–	0,00 €
Software (código abierto)	–	–	–	0,00 €
COSTE TOTAL DE RECURSOS MATERIALES				373,33 €
GASTOS GENERALES (costes indirectos)	15 %	sobre CD		866,00 €
BENEFICIO INDUSTRIAL	6 %	sobre CD + CI		398,36 €
MATERIAL FUNGIBLE				
Impresión				100,00 €
Encuadernación				300,00 €
SUBTOTAL PRESUPUESTO				7.437,69 €
IVA APLICABLE			21 %	1.561,92 €
TOTAL PRESUPUESTO				8.999,61 €